upvoted 1 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Selected Answer: A

Option A. Edit the job to use job bookmarks.

Job bookmarks in AWS Glue allow the ETL job to track the data that has been processed and to skip data that has already been processed. This can prevent AWS Glue from reprocessing old data and can improve the performance of the ETL job by only processing new data. To use job bookmarks, the solutions architect can edit the job and set the "Use job bookmark" option to "True". The ETL job will then use the job bookmark to track the data that has been processed and skip data that has already been processed in subsequent runs.

upvoted 3 times

career360guru 1 year, 12 months ago

Selected Answer: A

Option A

upvoted 1 times

SilentMilli 2 years ago

Selected Answer: A

It's obviously A. Bookmarks serve this purpose

upvoted 1 times

Wpcorgan 2 years ago

A is correct

upvoted 2 times

LeGloupier 2 years, 1 month ago

Selected Answer: A

A
https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #104 *Topic 1*

A solutions architect must design a highly available infrastructure for a website. The website is powered by Windows web servers that run on Amazon EC2 instances. The solutions architect must implement a solution that can mitigate a large-scale DDoS attack that originates from thousands of IP addresses. Downtime is not acceptable for the website.
Which actions should the solutions architect take to protect the website from such an attack? (Choose two.)

A. Use AWS Shield Advanced to stop the DDoS attack. **Most Voted**

B. Configure Amazon GuardDuty to automatically block the attackers.

C. Configure the website to use Amazon CloudFront for both static and dynamic content. **Most Voted**

D. Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs.

E. Use EC2 Spot Instances in an Auto Scaling group with a target tracking scaling policy that is set to 80% CPU utilization.

Correct Answer: *AC*

Community vote distribution

AC (89%) Other (11%)

Comments

alvarez100 Highly Voted 2 years, 8 months ago

Selected Answer: AC

I think it is AC, reason is they require a solution that is highly available. AWS Shield can handle the DDoS attacks. To make the solution HA you can use cloud front. AC seems to be the best answer imo.
AB seem like redundant answers. How do those answers make the solution HA?

upvoted 26 times

attila9778 2 years, 6 months ago

A - AWS Shield Advanced
C - (protecting this option) IMO: AWS Shield Advanced has to be attached. But it can not be attached directly to EC2 instances.
According to the docs: https://aws.amazon.com/shield/
It requires to be attached to services such as CloudFront, Route 53, Global Accelerator, ELB or (in the most direct way using) Elastic IP (attached to the EC2 instance)

upvoted 33 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: AC

Option A. Use AWS Shield Advanced to stop the DDoS attack.

It provides always-on protection for Amazon EC2 instances, Elastic Load Balancers, and Amazon Route 53 resources. By using AWS Shield Advanced, the solutions architect can help protect the website from large-scale DDoS attacks.

Option C. Configure the website to use Amazon CloudFront for both static and dynamic content.

CloudFront is a content delivery network (CDN) that integrates with other Amazon Web Services products, such as Amazon S3 and Amazon EC2, to deliver content to users with low latency and high data transfer speeds. By using CloudFront, the solutions architect can distribute the website's content across multiple edge locations, which can help absorb the impact of a DDoS attack and reduce the risk of downtime for the website.

upvoted 23 times

satyaammm Most Recent 5 months, 1 week ago

Selected Answer: AC

AWS Shield Advanced protects form DDos attacks while CloudFront deals with the downtime.

upvoted 1 times

XXXXXlNN 9 months ago

Note great options for us to select but AC seem make more sense comparing to others

upvoted 1 times

KTEgghead 10 months, 2 weeks ago

Selected Answer: AC

CoPilot - "No, you do not need Amazon CloudFront to implement AWS Shield Advanced. AWS Shield Advanced provides protection for several AWS services, including Amazon EC2, Elastic Load Balancing (ELB), AWS Global Accelerator, and Amazon Route 53 resources, in addition to CloudFront distributions1. It's designed to offer more sophisticated protection against Distributed Denial of Service (DDoS) attacks, regardless of the AWS service being used1. However, it's important to note that while CloudFront is not a requirement, using AWS Shield Advanced with CloudFront can enhance your application's security by providing additional DDoS protection."

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: AC

A and C is the most logical combination, we implement cloudfront so we can use shield advanced. Both of these options mitigate the impact of a DDOS attack.

upvoted 2 times

jatric 11 months, 2 weeks ago

Selected Answer: AC

AC is more close to meet the requirenment

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: AC

A: For DDoS attakcs
C: For scalable available site

B: Irrelevant
D: How would Lambda identify the attacker IP even if this was possible (ACL has a limit of 40 rules each way)
E: Scaling is not an issue here

upvoted 5 times

xdkonorek2 1 year, 7 months ago

Selected Answer: AC

A - use aws shield advanced for DDoS protection, but it cannot be used with EC2 instace if it's not using EIP, which is not mentioned
C - but it can be used with cloudfront distribution
thus AC is the answer

upvoted 3 times

Ruffyit 1 year, 7 months ago

DDoS attack will choose the AWS Shield Advanced
Cloudfront have attached the WAF
upvoted 2 times

Devsin2000 1 year, 8 months ago

Selected Answer: AE

A - no brainer
E = "must design a highly available infrastructure". I am not sure if CloudFront addresses this requirement.
upvoted 1 times

pentium75 1 year, 5 months ago

Is CloudFront not HA? Answer E uses Spot instances which might be unavailable, thus are NEVER an option for HA.
upvoted 4 times

sidharthwader 1 year, 3 months ago

You are right if it was On demand instances we could think of E
upvoted 2 times

LoXoL 1 year, 5 months ago

pentium75 is right.
upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: AC

Mitigate a large-scale DDoS attack = AWS Shield Advanced
Downtime is not acceptable for the website = high availability = Amazon CloudFront
upvoted 3 times

mtmayer 1 year, 9 months ago

Selected Answer: D

yeah , AWS Shield Advanced can be used directly on EC2.....
https://docs.aws.amazon.com/waf/latest/developerguide/ddos-protections-by-resource-type.html
upvoted 1 times

pentium75 1 year, 5 months ago

Why D then?
upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AC

Cloud front supports SHIELD ADVANCED integration
upvoted 3 times

diabloexodia 1 year, 11 months ago

Cloud front supports SHIELD ADVANCED integration
upvoted 2 times

Aash24 1 year, 11 months ago

Selected Answer: D

D should be the one here
upvoted 3 times

pentium75 1 year, 5 months ago

"Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs"?????

Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs ?????

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: AC

A. AWS Shield Advanced provides advanced DDoS protection for AWS resources, including EC2. It includes features such as real-time threat intelligence, automatic protection, and DDoS cost protection.

C. CloudFront is a CDN service that can help mitigate DDoS attacks. By routing traffic through CloudFront, requests to the website are distributed across multiple edge locations, which can absorb and mitigate DDoS attacks more effectively. CloudFront also provides additional DDoS protection features, such as rate limiting, SSL/TLS termination, and custom security policies.

B. While GuardDuty can detect and provide insights into potential malicious activity, it is not specifically designed for DDoS mitigation.

D. Network ACLs are not designed to handle high-volume traffic or DDoS attacks efficiently.

E. Spot Instances are a cost optimization strategy and may not provide the necessary availability and protection against DDoS attacks compared to using dedicated instances with DDoS protection mechanisms like Shield Advanced and CloudFront.

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #105 *Topic 1*

A company is preparing to deploy a new serverless workload. A solutions architect must use the principle of least privilege to configure permissions that will be used to run an AWS Lambda function. An Amazon EventBridge (Amazon CloudWatch Events) rule will invoke the function.
Which solution meets these requirements?

A. Add an execution role to the function with lambda:InvokeFunction as the action and * as the principal.

B. Add an execution role to the function with lambda:InvokeFunction as the action and Service: lambda.amazonaws.com as the principal.

C. Add a resource-based policy to the function with lambda:* as the action and Service: events.amazonaws.com as the principal.

D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (97%) B

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D

Best way to check it... The question is taken from the example shown here in the documentation:
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-use-resource-based.html#eb-lambda-permissions

upvoted 38 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: D

The correct solution is D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal.

The principle of least privilege requires that permissions are granted only to the minimum necessary to perform a task. In this case, the Lambda function needs to be able to be invoked by Amazon EventBridge (Amazon CloudWatch Events). To meet these requirements, you can add a resource-based policy to the function that allows the InvokeFunction action to be performed by the Service: events.amazonaws.com principal. This will allow Amazon EventBridge to invoke the function, but will not grant any additional permissions to the function.

upvoted 29 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Why other options are wrong

Option A is incorrect because it grants the lambda:InvokeFunction action to any principal (*), which would allow any entity to invoke the function and goes beyond the minimum permissions needed.

Option B is incorrect because it grants the lambda:InvokeFunction action to the Service: lambda.amazonaws.com principal, which would allow any Lambda function to invoke the function and goes beyond the minimum permissions needed.

Option C is incorrect because it grants the lambda:* action to the Service: events.amazonaws.com principal, which would allow Amazon EventBridge to perform any action on the function and goes beyond the minimum permissions needed.

upvoted 23 times

huaze_lei Most Recent 9 months, 1 week ago

Selected Answer: D

Following the principle of least privilege, you should not grant Events the * privilege. Just enough to perform its job will do.

Also, you need a resource-based policy to attach to the function, for Events to be able to execute the function

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D is the correct answer

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

This is a good example article with nice learning material.
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-run-lambda-schedule.html

upvoted 2 times

chasingsummer 1 year, 6 months ago

Selected Answer: D

Good explanation from ChatGPT:

In order to adhere to the principle of least privilege when configuring permissions for an AWS Lambda function invoked by an Amazon EventBridge (CloudWatch Events) rule, the most appropriate solution would be:

D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal.

This solution involves attaching a resource-based policy to the Lambda function. It specifies that the only entity allowed to invoke the Lambda function is the Amazon EventBridge service (represented by the principal events.amazonaws.com) and restricts the action to only invoking the function (lambda:InvokeFunction). This aligns with the principle of least privilege by granting the necessary permissions explicitly to the service that needs them, without providing overly permissive access.

upvoted 3 times

MiniYang 1 year, 6 months ago

Selected Answer: B

Is anyone can explain why B is can't be a good choice? The option adds the execution role to the function, with lambda:InvokeFunction as the action and Service: lambda.amazonaws.com as the body. This restricts the Lambda function to only the Lambda service, providing an effective layer of security. and fully complies with the principle of least privilege

upvoted 3 times

pentium75 1 year, 5 months ago

Because the question is about the permission 'to run the function' (permission for the administrator to invoke it), while B is about execution permissions (permission for the function to access resources).

upvoted 1 times

Evonne_HY 1 year, 9 months ago

why not choose B, an execution role is attached to lambda and a policy is attached to an execution role
upvoted 1 times

Georgeyp 1 year, 8 months ago

B would be the wrong choice as the both roles are granted to lambda, however the question requires Eventbridge to call the Lambda function.
upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

lambda:InvokeFunction is the action needed to invoke the Lambda function.
Service: events.amazonaws.com is the principal (the AWS service) that is allowed to invoke the Lambda function. In this case, you're explicitly allowing CloudWatch Events to invoke the function.
upvoted 2 times

MNotABot 1 year, 11 months ago

D
* is BIG NO. And we are talking about policy --> hence D
upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: D

In this solution, a resource-based policy is added to the Lambda function, which allows the specified principal (events.amazonaws.com) to invoke the function. The lambda:InvokeFunction action provides the necessary permission for the Amazon EventBridge rule to trigger the Lambda function.

Option A is incorrect because it assigns the lambda:InvokeFunction action to all principals (*), which grants permission to invoke the function to any entity, which is broader than necessary.

Option B is incorrect because it assigns the lambda:InvokeFunction action to the specific principal "lambda.amazonaws.com," which is the service principal for AWS Lambda. However, the requirement is for the EventBridge service principal to invoke the function.

Option C is incorrect because it assigns the lambda:* action to the specific principal "events.amazonaws.com," which is the service principal for Amazon EventBridge. However, it grants broader permissions than necessary, allowing any Lambda function action, not just lambda:InvokeFunction.
upvoted 4 times

Abrar2022 2 years ago

Option C is incorrect, the reason is that, firstly, lambda:* allows Amazon EventBridge to perform any action on the function and this is beyond the minimum permissions needed.
upvoted 2 times

Rahulbit34 2 years, 1 month ago

Since its for Lamda which is a resource, resource policy is the trick
upvoted 2 times

bdp123 2 years, 4 months ago

Selected Answer: D

https://docs.aws.amazon.com/eventbridge/latest/userguide/resource-based-policies-eventbridge.html#lambda-permissions
upvoted 1 times

gustavtd 2 years, 5 months ago

Selected Answer: D

The definition scope of D is the smallest, so is it
upvoted 1 times

techhb 2 years, 5 months ago

Selected Answer: D

events.amazonaws.com is principal for eventbridge

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #106 *Topic 1*

A company is preparing to store confidential data in Amazon S3. For compliance reasons, the data must be encrypted at rest. Encryption key usage must be logged for auditing purposes. Keys must be rotated every year.
Which solution meets these requirements and is the MOST operationally efficient?

A. Server-side encryption with customer-provided keys (SSE-C)

B. Server-side encryption with Amazon S3 managed keys (SSE-S3)

C. Server-side encryption with AWS KMS keys (SSE-KMS) with manual rotation

D. Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation **Most Voted**

Correct Answer: *D*

Community vote distribution

D (93%) B (7%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D

The MOST operationally efficient one is D.
Automating the key rotation is the most efficient.
Just to confirm, the A and B options don't allow automate the rotation as explained here:
https://aws.amazon.com/kms/faqs/#:~:text=You%20can%20choose%20to%20have%20AWS%20KMS%20automatically%20rotate%20KMS,KMS%20custom%20key%20store%20feature

upvoted 22 times

vadiminski_a 2 years, 5 months ago

In addition you cannot log key usage in B, for A I am not certain

upvoted 2 times

ocbn3wby 2 years, 6 months ago

Thank you for the explanation.

upvoted 1 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: D

SSE-KMS provides a secure and efficient way to encrypt data at rest in S3. SSE-KMS uses KMS to manage the encryption keys securely. With SSE-KMS, encryption keys can be automatically rotated using KMS key rotation feature, which simplifies the key management process and ensures compliance with the requirement to rotate keys every year.

Additionally, SSE-KMS provides built-in audit logging for encryption key usage through CloudTrail, which captures API calls related to the management and usage of KMS keys. This meets the requirement for logging key usage for auditing purposes.

Option A (SSE-C) requires customers to provide their own encryption keys, but it does not provide key rotation or built-in logging of key usage.
Option B (SSE-S3) uses Amazon S3 managed keys for encryption, which simplifies key management but does not provide key rotation or detailed key usage logging.
Option C (SSE-KMS with manual rotation) uses AWS KMS keys but requires manual rotation, which is less operationally efficient than the automatic key rotation available with option D.

upvoted 10 times

satyaammm Most Recent 5 months, 1 week ago

Selected Answer: D

Automating the key rotation is the most important here and hence D is the most suitable option here.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans D - just the Amazon provided service with key automatic key rotation

upvoted 1 times

MehulKapadia 1 year, 2 months ago

Selected Answer: D

Correct Answer: D
Automatic Key Rotation = KMS, hence Option A & B are not correct answer.
Hence Possible answer is Option C or D. Now mentioned in the requirement that key rotation solution must be automated. So Option C is not the correct answer.
Correct Answer: D - SSE with KMS which support automatic key rotation.

upvoted 2 times

Karun3294 1 year, 3 months ago

I got this question in exam today (FEB 21, 2024)

upvoted 5 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

I'll go for D as SSS-S3 has unpublished scheduled of rotation which may or may not be "each year".
https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingKMSEncryption.html

upvoted 2 times

rcptryk 1 year, 6 months ago

Selected Answer: B

SSE-S3 can be used for logging in cloudtrail since January 5, 2023
https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingServerSideEncryption.html

upvoted 2 times

pentium75 1 year, 5 months ago

But "keys must be rotated every year". I understand that SSE-S3 rotates the keys "regularly" but you have no influence on the schedule.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The correct answer is D. Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation.

SSE-KMS is the most secure way to encrypt data in Amazon S3. It uses AWS KMS, which is a highly secure key management service that is managed by AWS. AWS KMS logs all key usage, so the company can meet its compliance requirements. AWS KMS also rotates keys automatically, so the company does not have to worry about manually rotating keys.

upvoted 4 times

SilentMilli 2 years, 5 months ago

Selected Answer: D

Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation meets the requirements and is the most operationally efficient solution. This option allows you to use AWS KMS to automatically rotate the keys every year, which simplifies key management. In addition, key usage is logged for auditing purposes, and the data is encrypted at rest to meet compliance requirements.

upvoted 3 times

Zerotn3 2 years, 5 months ago

Selected Answer: B

mazon API Gateway is a fully managed service that makes it easy to create, publish, maintain, monitor, and secure APIs at any scale. You can use API Gateway to create a REST API that exposes the location data as an API endpoint, allowing you to access the data from your analytics platform.

AWS Lambda is a serverless compute service that lets you run code in response to events or HTTP requests. You can use Lambda to write the code that retrieves the location data from your data store and returns it to API Gateway as a response to API requests. This allows you to scale the API to handle a large number of requests without the need to provision or manage any infrastructure.

upvoted 2 times

pentium75 1 year, 5 months ago

This question is about server-side encryption, not API Gateway

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: D

The most operationally efficient solution that meets the requirements listed would be option D: Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation.

SSE-KMS allows you to use keys that are managed by the AWS Key Management Service (KMS) to encrypt your data at rest. KMS is a fully managed service that makes it easy to create and control the encryption keys used to encrypt your data. With automatic key rotation enabled, KMS will automatically create a new key for you on a regular basis, typically every year, and use it to encrypt your data. This simplifies the key rotation process and reduces the operational burden on your team.

In addition, SSE-KMS provides logging of key usage through AWS CloudTrail, which can be used for auditing purposes.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Why other options are wrong

Option A: Server-side encryption with customer-provided keys (SSE-C) would require you to manage the encryption keys yourself, which can be more operationally burdensome.

Option B: Server-side encryption with Amazon S3 managed keys (SSE-S3) does not allow for key rotation or logging of the key usage.

Option C: Server-side encryption with AWS KMS keys (SSE-KMS) with manual rotation would require you to manually initiate the key rotation process, which can be more operationally burdensome compared to automatic rotation.

upvoted 4 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

Berny 2 years, 6 months ago

You can choose to have AWS KMS automatically rotate KMS keys every year, provided that those keys were generated within AWS KMS HSMs. Automatic key rotation is not supported for imported keys, asymmetric keys, or keys generated in a

CloudHSM cluster using the AWS KMS custom key store feature. If you choose to import keys to AWS KMS or asymmetric keys or use a custom key store, you can manually rotate them by creating a new KMS key and mapping an existing key alias from the old KMS key to the new KMS key.

upvoted 2 times

PavelTech 2 years, 6 months ago

Can anybody correct me if I'm wrong, KMS does not offer automatic rotations but SSE-KMS only allows automatic rotation once in 3 years thus if we want rotation every year we need to rotate it manually?

upvoted 2 times

JayBee65 2 years, 5 months ago

You're wrong :) "All AWS managed keys are automatically rotated every year. You cannot change this rotation schedule." https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#customer-cmk

upvoted 2 times

PS_R 2 years, 7 months ago

Selected Answer: D

Agree Also, SSE-S3 cannot be audited.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #107 *Topic 1*

A bicycle sharing company is developing a multi-tier architecture to track the location of its bicycles during peak operating hours. The company wants to use these data points in its existing analytics platform. A solutions architect must determine the most viable multi-tier option to support this architecture. The data points must be accessible from the REST API.
Which action meets these requirements for storing and retrieving location data?

A. Use Amazon Athena with Amazon S3.

B. Use Amazon API Gateway with AWS Lambda.

C. Use Amazon QuickSight with Amazon Redshift.

D. Use Amazon API Gateway with Amazon Kinesis Data Analytics. Most Voted

Correct Answer: *D*

Community vote distribution

D (47%) B (46%) A (6%)

Comments

ArielSchivo Highly Voted 2 years, 7 months ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
“The company wants to use these data points in its existing analytics platform” so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.

upvoted 105 times

Tsige 8 months ago

While API Gateway and Lambda can work together for processing requests, this setup doesn't efficiently handle continuous real-time data streaming and analysis, which is required for tracking the bicycle locations. Lambda is stateless and is better suited for on-demand execution, not for real-time analytics of streaming data. So, the correct option is D

upvoted 9 times

alfonso_ciampa 1 year, 11 months ago

You are right, but it clearly say "store data".
AWS Lambda don't store data, Kinesis could.

upvoted 11 times

FlyingHawk 4 months, 3 weeks ago

AWS Lambda provides a serverless way to process incoming requests and interact with a storage backend.
This solution can store location data in a service like Amazon DynamoDB or Amazon S3 and use Lambda functions to retrieve and serve this data through the API Gateway. This is cost-effective, scalable, and straightforward for real-time queries.

upvoted 2 times

MutiverseAgent 1 year, 11 months ago

B might work but D works better. B requieres API gateway + lambda for data input & output, whereas D is a broader solution, as Kinesis Data Analytics APIs can be used to extract and process data better that API Gateway + Lambdas. Also, Kinesis is highly recommended for telemetry data which is the question scenario. @See Kinesys flexible API (https://aws.amazon.com/documentation-overview/kinesis-data-analytics/)

upvoted 12 times

MutiverseAgent 1 year, 11 months ago

Also by using kinesis the analytics platform will have a storing buffer to take & process data through the kinesys API. The lambda aproach in the B scenario is to wide and leaves many loose ends.

upvoted 5 times

bullrem 2 years, 4 months ago

AWS Lambda is a serverless compute service that can be used to run code in response to specific events, such as changes to data in an Amazon S3 bucket or updates to a DynamoDB table. It could be used to process the location data, but it doesn't provide storage solution. Therefore, it would not be the best option for storing and retrieving location data in this scenario.

upvoted 13 times

Six_Fingered_Jose Highly Voted 2 years, 7 months ago

Selected Answer: D

I dont understand why you will vote B?
how are you going to store data with just lambda?
> Which action meets these requirements for storing and retrieving location data

In this use case there will obviously be a ton of data and you want to get real-time location data of the bicycles, and to analyze all these info kinesis is the one that makes most sense here.

upvoted 79 times

a070112 2 years, 5 months ago

Lambda isn't storing the data themselves. It's triggering the data store to the company's "existing data analytics platform"

upvoted 10 times

kmliuy73 2 years, 6 months ago

Real-time analytics on Kinesis Data Streams & Firehose using SQL, not store db ...

upvoted 5 times

Six_Fingered_Jose 2 years, 7 months ago

https://aws.amazon.com/blogs/aws/real-time-hotspot-detection-in-amazon-kinesis-analytics/

upvoted 5 times

UWSFish 2 years, 7 months ago

I don't think you need to worry about storing data. The question states their is an existing platform.

upvoted 4 times

TEC65 Most Recent 1 month, 4 weeks ago

Selected Answer: D

better for streaming data

upvoted 1 times

Vivobook11 2 months, 2 weeks ago

Selected Answer: D

D. Use Amazon API Gateway with Amazon Kinesis Data Analytics.
Why?
Real-time Data Processing: Amazon Kinesis Data Analytics can process streaming data (e.g., bicycle locations) in real-time.

REST API Access: Amazon API Gateway can provide a secure REST API for accessing the processed data.

Scalability & Performance: Kinesis efficiently ingests large volumes of data during peak hours.

Integration with Analytics: The processed data can be stored in Amazon S3, DynamoDB, or Redshift for further analytics.
upvoted 2 times

Faraz999 2 months, 2 weeks ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
“The company wants to use these data points in its existing analytics platform” so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.
upvoted 1 times

CloudExpert01 2 months, 2 weeks ago

Selected Answer: B

when choosing between B and D:
"The company wants to use these data points in its existing analytics platform." so need for Lambda to integrate data points in its existing analytics platform
upvoted 1 times

Faraz999 2 months, 2 weeks ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
“The company wants to use these data points in its existing analytics platform” so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.
upvoted 1 times

Faraz999 2 months, 2 weeks ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
“The company wants to use these data points in its existing analytics platform” so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.
upvoted 1 times

Saloniip 3 months, 3 weeks ago

Selected Answer: B

lambda can store data in db which is not mentioned but can be implemented, wherein kinesis data analytics would be more complex to implement just for simple location tracking. Also, a data analytics tool is already there. Hence, B is more suitable.
upvoted 1 times

tch 4 months, 1 week ago

Selected Answer: D

Amazon Kinesis Data Analytics allows you to build more complex analytics applications that support flexible processing choices and robust fault-tolerance with exactly-once processing without duplicates, and analytics that can be performed over an entire data stream across multiple logical partitions. With KDA, you can analyze data over multiple types of aggregation windows (tumbling window, stagger window, sliding window, session window) using either the event time or the processing tim
upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: D

Lambda is not most viable, or real time
upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: B

Given that the company already has an existing analytics platform and needs a solution for storing and retrieving location data that can be accessed through a REST API, the best choice is:

B. Use Amazon API Gateway with AWS Lambda

upvoted 3 times

FlyingHawk 4 months, 3 weeks ago

D. Amazon API Gateway with Amazon Kinesis Data Analytics: Kinesis Data Analytics is for continuous real-time stream processing. If the requirement focuses only on storing and retrieving (not streaming) data, this adds unnecessary complexity.

upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: D

Kinesis is specifically designed for telemetry data, making it a better fit for tracking bicycle locations.

upvoted 3 times

FlyingHawk 4 months, 3 weeks ago

I changed my mind to B now as there is existing analytic platform already, the data will be stored in the existing analytic platform, so you can use Lambda to process and store data in the existing analytic platform.

upvoted 1 times

Rcosmos 5 months ago

Selected Answer: D

B. Amazon API Gateway com AWS Lambda:

Essa combinação é útil para fornecer APIs REST, mas o Lambda não é otimizado para ingestão contínua de dados de localização em tempo real. Ele seria mais adequado para processos eventuais ou tarefas específicas.

upvoted 2 times

hashepsut 5 months, 2 weeks ago

Selected Answer: B

the best answer is B: Use Amazon API Gateway with AWS Lambda

Reasoning:

API Gateway provides the required REST API endpoint
Lambda can handle real-time processing of location data
This combination is cost-effective and scalable
Can easily integrate with other AWS services for analytics
Provides the necessary multi-tier architecture (API Layer, Processing Layer, Storage Layer)

upvoted 1 times

EllenLiu 6 months ago

Selected Answer: D

A common use is the real-time aggregation of data followed by loading the aggregate data into a data warehouse or map-reduce cluster.
https://docs.aws.amazon.com/streams/latest/dev/introduction.html#using-the-service

upvoted 1 times

kimm_10 6 months ago

Selected Answer: D

D is the most viable multi-tier option, as it supports real-time data ingestion, analytics, and accessibility through a REST API.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #108 *Topic 1*

A company has an automobile sales website that stores its listings in a database on Amazon RDS. When an automobile is sold, the listing needs to be removed from the website and the data must be sent to multiple target systems.
Which design should a solutions architect recommend?

A. Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) queue for the targets to consume. **Most Voted**

B. Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) FIFO queue for the targets to consume.

C. Subscribe to an RDS event notification and send an Amazon Simple Queue Service (Amazon SQS) queue fanned out to multiple Amazon Simple Notification Service (Amazon SNS) topics. Use AWS Lambda functions to update the targets.

D. Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

Correct Answer: *A*

Community vote distribution

A (50%) D (45%) Other

Comments

romko Highly Voted 2 years, 6 months ago

Selected Answer: A

Interesting point that Amazon RDS event notification doesn't support any notification when data inside DB is updated.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_Events.overview.html
So subscription to RDS events doesn't give any value for Fanout = SNS => SQS

B is out because FIFO is not required here.

A is left as correct answer

upvoted 90 times

Tsige 8 months ago

Option A suggests creating an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) queue for the targets to consume. While this approach can work, it has a few limitations compared to Option D:

Scalability and Fan-out: Option A uses a single SQS queue, which means all target systems would need to poll the same queue. This can become a bottleneck if multiple systems need to process the data simultaneously. Option D, on the other hand, uses an SNS topic to fan out the event to multiple SQS queues, allowing each target system to have its own queue. This improves scalability and ensures that each target system can process the data independently.
Option D offers a more scalable, decoupled, and flexible solution for handling the event notifications and distributing the data to multiple target systems.

upvoted 4 times

mknarula 11 months, 3 weeks ago

But... SQS is a queue and is incapable of sending messages to "multiple target systems". SNS is pub/sub and topics can be subscribed by multiple apps to update when such an even occurs. Moreover Amazon RDS uses native capabilities of DBs like Postgres, MS SQL for change data capture. This can be used to send notifications to SNS

upvoted 4 times

nauman001 2 years, 2 months ago

Listing the Amazon RDS event notification categories.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_Events.ListingCategories.html:

upvoted 3 times

Jiang_aws1 2 years, 6 months ago

D is connect
RDS event notification by RDS stream or advance audit DML so it is possible

upvoted 2 times

Jiang_aws1 2 years, 6 months ago

The key is "Fanned out" due to "Multiple target systems" need to update

upvoted 4 times

JayBee65 2 years, 5 months ago

Please provide reference for this claim: " event notification by RDS stream or advance audit DML"

upvoted 4 times

ksolovyov Highly Voted 2 years, 5 months ago

Selected Answer: A

RDS events only provide operational events such as DB instance events, DB parameter group events, DB security group events, and DB snapshot events. What we need in the scenario is to capture data-modifying events (INSERT, DELETE, UPDATE) which can be achieved thru native functions or stored procedures.

upvoted 15 times

BlueVolcano1 2 years, 4 months ago

I agree with it requiring a native function or stored procedure, but can they in turn invoke a Lambda function? I have only seen this being possible with Aurora, but not RDS - and I'm not able to find anything googling for it either. I guess it has to be possible, since there's no other option that fits either.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Integrating.Lambda.html

upvoted 1 times

BlueVolcano1 2 years, 4 months ago

To add to that though, A also states to only use SQS (no SNS to SQS fan-out), which doesn't seem right as the message needs to go to multiple targets?

upvoted 13 times

bora4motion Most Recent 2 weeks, 2 days ago

Selected Answer: D

going with d as you need sns to fan out to multiple systems

upvoted 1 times

Kp002 1 month ago

Selected Answer: D

Why not A or B: Lambda cannot be directly triggered by Amazon RDS data changes unless you're using Amazon Aurora with the advanced Aurora MySQL features, which support Aurora database triggers to invoke Lambda. If it's standard RDS, you'd need to build this logic into the application itself or poll the database — not ideal.
• FIFO queues (Option B) are only needed if message order is critical, which isn't indicated in the question.
upvoted 1 times

c6ee0dd 1 month, 3 weeks ago

Selected Answer: D

D : event publishes message to a topic making it available to multiple consumers
A : Its a single SQS queue which can lead to problems when multiple consumers are consuming the data concurrently
upvoted 1 times

TEC65 1 month, 4 weeks ago

Selected Answer: D

D - RDS cannot send notification to lambda
upvoted 1 times

network_enthusiast 2 months ago

Selected Answer: B

B is the correct answer
upvoted 1 times

CloudExpert01 2 months, 2 weeks ago

Selected Answer: A

RDS event notifications won't capture row-level changes directly, and adding complexity with SQS and SNS may be unnecessary.
upvoted 1 times

ChhatwaniB 3 months, 1 week ago

Selected Answer: D

SQS won't work for multiple targets. Once a message is consumed by a target it will be removed from the queue
upvoted 3 times

Saloniip 3 months, 3 weeks ago

Selected Answer: D

We need to send same msg to multiple targets that a data bas been removed. Not different msgs to different targets. so option d is much better solution.
upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: D

A is not right because one SQS consumed by multiple targets with different process is not possible. Ideally it should be sent to multiple SQS. D is working ok.
upvoted 4 times

CristiaNNN 4 months, 3 weeks ago

Selected Answer: D

Option D: Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

Correct Choice: This approach leverages RDS event notifications and SNS for fanning out updates to multiple SQS queues. Each target can then independently process updates using Lambda functions, making it scalable and modular.
upvoted 4 times

Rcosmos 5 months ago

Selected Answer: D

Conclusão:
A opção D (SNS para distribuição e SQS para entrega confiável) é a solução mais adequada, escalável e resiliente para lidar

A opção D (SNS para distribuição e SQS para entrega confiável) é a solução mais adequada, escalável e resiliente para lidar com a necessidade de envio de dados a vários sistemas de destino após a venda de um automóvel.
upvoted 1 times

4729e6c 5 months ago

Selected Answer: D

Option D is the best choice because it uses RDS event notifications to trigger an SNS topic that fans out to multiple SQS queues, enabling each target system to process events independently and reliably. This approach ensures scalability, decoupling, and durable message storage. Option A is not ideal because RDS does not natively trigger Lambda functions; an external polling mechanism would be required, increasing complexity and operational overhead.
upvoted 2 times

Mischi 5 months, 3 weeks ago

Selected Answer: D

The combination of Amazon SNS and Amazon SQS together with AWS Lambda provides a robust, scalable and efficient solution for distributing RDS events to multiple target systems. This architecture ensures that data is transmitted reliably and that each target system can process the information independently, optimally meeting business requirements.
upvoted 2 times

dipenich 5 months, 4 weeks ago

Selected Answer: D

The requirement is to remove automobile listings from the website after a sale and send the data to multiple target systems in a decoupled, reliable, and scalable manner. Here's how Option D meets the requirements:

Event Notification from RDS:
Amazon RDS can publish event notifications to Amazon SNS topics. When a record is updated (e.g., when a sale is recorded), this triggers an event.

Fan-out to Multiple SQS Queues:
The SNS topic fans out the event notification to multiple SQS queues, allowing parallel processing for different target systems. Each target system consumes messages from its respective queue, ensuring reliable and independent data delivery.

AWS Lambda for Target Updates:
Lambda functions subscribed to the SQS queues process the messages and perform updates in the target systems.

Decoupled Design:
The use of SNS for fan-out and SQS for message queuing decouples the systems. This ensures that issues with one target system do not impact the others.
upvoted 3 times

EllenLiu 6 months ago

Selected Answer: D

key point: Even though lambda is integrated with RDS, it is not best practices for multiple consumers.
SNS + SQS Fan-Out should be chosen
https://aws.amazon.com/getting-started/hands-on/send-fanout-event-notifications/?nc1=h_ls
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #109 *Topic 1*

A company needs to store data in Amazon S3 and must prevent the data from being changed. The company wants new objects that are uploaded to Amazon S3 to remain unchangeable for a nonspecific amount of time until the company decides to modify the objects. Only specific users in the company's AWS account can have the ability 10 delete the objects.
What should a solutions architect do to meet these requirements?

A. Create an S3 Glacier vault. Apply a write-once, read-many (WORM) vault lock policy to the objects.

B. Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Set a retention period of 100 years. Use governance mode as the S3 bucket's default retention mode for new objects.

C. Create an S3 bucket. Use AWS CloudTrail to track any S3 API events that modify the objects. Upon notification, restore the modified objects from any backup versions that the company has.

D. Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Add a legal hold to the objects. Add the s3:PutObjectLegalHold permission to the IAM policies of users who need to delete the objects. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (78%) B (22%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D

A - No as "specific users can delete"
B - No as "nonspecific amount of time"
C - No as "prevent the data from being change"
D - The answer: "The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed."
https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-legal-hold.html

upvoted 38 times

PassNow1234 2 years, 5 months ago

The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.

Correct
upvoted 2 times

Chunsli Highly Voted 2 years, 8 months ago

typo -- 10 delete the objects => TO delete the objects
upvoted 19 times

oddnoises 1 year, 8 months ago

they were trying to speak in binary lol
upvoted 7 times

reviewmine 1 year, 3 months ago

HAHAHA
upvoted 1 times

jorgemenegaz Most Recent 2 weeks ago

Selected Answer: B

Here's why option B is suitable:

S3 Object Lock: This feature allows you to prevent objects from being deleted or changed for a specified retention period. In governance mode, even if a user has delete permissions, they cannot delete the objects until the retention period has expired.

Versioning: Enabling versioning ensures that multiple versions of an object are stored. This provides an additional layer of data protection, allowing you to restore data if an object is accidentally modified or deleted.

Retention Period: By setting a long retention period (like 100 years), you ensure compliance with data retention policies beyond short-term needs.
upvoted 1 times

c12ab95 4 weeks ago

Selected Answer: B

Why This Works
S3 Object Lock with Governance Mode:

Immutable by Default: New objects inherit a 100-year retention period, ensuring they cannot be modified or deleted during this time.

Flexible Override: Authorized users (with s3:BypassGovernanceRetention permissions) can shorten the retention period or delete objects before 100 years, aligning with the requirement for nonspecific retention until the company decides.

Versioning:

Protects against accidental overwrites or deletions by maintaining multiple object versions.

Governance Mode Benefits:

Allows controlled exceptions for authorized users while blocking unauthorized changes.

Why Other Options Fail
Option Shortcoming
A Glacier vaults are for archival, not active S3 storage.
C CloudTrail is reactive (audits changes but doesn't prevent them).
D Legal holds are indefinite but require manual application per object, which is not scalable for new uploads.
upvoted 1 times

CristiaNNN 4 months, 3 weeks ago

Selected Answer: D

Option D: Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

Correct Choice: This approach leverages RDS event notifications and SNS for fanning out updates to multiple SQS queues. Each target can then independently process updates using Lambda functions, making it scalable and modular.
upvoted 1 times

upvoted 1 times

satyaammm 5 months, 1 week ago

Selected Answer: D

S3 Bucket Versioning helps deal with modifications while the legal hold coveys no deletes or changes until removed.

upvoted 1 times

aatikah 5 months, 4 weeks ago

Selected Answer: B

D is wrong
A legal hold can make objects immutable, but the question specifies new objects must remain unchangeable by default. Legal holds must be applied manually to individual objects, so they are not practical for this use case.

upvoted 1 times

PaulGa 9 months ago

Ans D - "The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed."
https://docs.amazonaws.cn/en_us/AmazonS3/latest/userguide/batch-ops-legal-hold.html
The other options do not make sense for the situation in hand.

upvoted 2 times

huaze_lei 9 months, 1 week ago

Selected Answer: D

The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.

You can use S3 Batch Operations with Object Lock to add legal holds to many Amazon S3 objects at once. You can do this by listing the target objects in your manifest and submitting that list to Batch Operations. Your S3 Batch Operations job with Object Lock legal hold runs until completion, until cancellation, or until a failure state is reached.

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D is the best choice

upvoted 3 times

professorx123 1 year, 1 month ago

Selected Answer: B

Adding legal holds to objects and managing permissions for users to delete objects does not provide the same level of data immutability and retention control as S3 Object Lock.

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html
A: WORM doesn't allow delete by some users
C: Irrelevant
D: Permission only allows putting legal hold on objects. Not a complete solution
B: Closest apart from 100 years as question is asking for indefinite. Governance allows modification by some users

upvoted 2 times

pentium75 1 year, 5 months ago

Selected Answer: B

"With governance mode, you protect objects against being deleted by most users, but you can still grant some users permission to alter the retention settings or delete the objects if necessary."

D is wrong because it applies Object Lock AND Legal Hold, which are two different things that achieve similar results. 'Adding the s3:PutObjectLegalHold permission' to user's policies would allow them to remove the Legal Hold but NOT the Object Lock. (Also, it would probably make more sense to add the permissions to the bucket policy, not the "IAM policies of users".)

upvoted 4 times

LoXoL 1 year, 5 months ago

Isn't Legal Hold a subcategory of Object Lock? Object Lock itself doesn't imply anything imho: you should go either for a Retention Mode OR Legal Hold. Why would you go for B if they ask "for a nonspecific amount of time"? Open to change my mind.

upvoted 1 times

Abitek007 1 year, 8 months ago

Selected Answer: D

I only picked this because of restricted users who can delete, and the easiest way of achieving this is them assuming the role

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

"The company wants new objects that are uploaded to Amazon S3 to remain unchangeable for a nonspecific amount of time until the company decides to modify the objects" = A legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.
s3:PutObjectLegalHold permission is required in your IAM role to add or remove legal hold from objects.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.

upvoted 2 times

RupeC 1 year, 10 months ago

Selected Answer: D

My understanding is that the s3:PutObjectLegalHold permission allows certain users to apply or remove the legal hold on objects in the S3 bucket. However, having the permission to apply or remove the legal hold does not necessarily mean users can override the hold set by another user.

Once the legal hold is set on an object, it is in effect until the hold is removed by the user who applied it or an admin with the necessary permissions. Other users, even if they have the s3:PutObjectLegalHold permission, won't be able to remove the hold unless they are granted access by the user who originally applied it.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #110 *Topic 1*

A social media company allows users to upload images to its website. The website runs on Amazon EC2 instances. During upload requests, the website resizes the images to a standard size and stores the resized images in Amazon S3. Users are experiencing slow upload requests to the website.
The company needs to reduce coupling within the application and improve website performance. A solutions architect must design the most operationally efficient process for image uploads.
Which combination of actions should the solutions architect take to meet these requirements? (Choose two.)

A. Configure the application to upload images to S3 Glacier.

B. Configure the web server to upload the original images to Amazon S3.

C. Configure the application to upload images directly from each user's browser to Amazon S3 through the use of a presigned URL **Most Voted**

D. Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded. Use the function to resize the image. **Most Voted**

E. Create an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function on a schedule to resize uploaded images.

Correct Answer: *CD*

Community vote distribution

CD (53%) BD (45%) Oth

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: CD

To meet the requirements of reducing coupling within the application and improving website performance, the solutions architect should consider taking the following actions:

C. Configure the application to upload images directly from each user's browser to Amazon S3 through the use of a pre-signed URL. This will allow the application to upload images directly to S3 without having to go through the web server, which can reduce the load on the web server and improve performance.

D. Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded. Use the function to resize the image. This will allow the application to resize images asynchronously, rather than having to do it synchronously during the upload request, which can improve performance.

upvoted 59 times

jdr75 2 years, 2 months ago

presigned URL is for download the data from S3, not for uploads, so the user does not upload anything. C is no correct.
upvoted 16 times

mauroicardi 1 year, 3 months ago

A user who does not have AWS credentials to upload a file can use a presigned URL to perform the upload.

https://boto3.amazonaws.com/v1/documentation/api/latest/guide/s3-presigned-urls.html
upvoted 7 times

Vandaman 3 months, 3 weeks ago

This makes sense. I will change my answers to CD
upvoted 1 times

EricYu2023 2 years, 1 month ago

Presigned URL can be use for upload.
upvoted 14 times

PoisonBlack 2 years, 1 month ago

https://docs.aws.amazon.com/AmazonS3/latest/userguide/PresignedUrlUploadObject.html
upvoted 9 times

AF_1221 2 years, 1 month ago

preassigned URL is for upload or download for temporary time and for specific users outside the company
upvoted 5 times

AF_1221 2 years, 1 month ago

but for temporary purpose not for permanent
upvoted 4 times

tuso 1 year, 4 months ago

So? You only need a presigned URL for the moment you upload the image, not forever
upvoted 2 times

AnhNguyen99 11 months ago

that's right
upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Why other options are wrong
Option A, Configuring the application to upload images to S3 Glacier, is not relevant to improving the performance of image uploads.

Option B, Configuring the webserver to upload the original images to Amazon S3, is not a recommended solution as it would not reduce coupling within the application or improve performance.

Option E, Creating an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function on a schedule to resize uploaded images, is not a recommended solution as it would not be able to resize images in a timely manner and would not improve performance.
upvoted 4 times

MutiverseAgent 1 year, 11 months ago

About your comments regarding option B)... But if images are being saved directly to S3 instead of the EBS/SSD storage of E2 instances as they originally were, the new approach will reduce coupling and improve performance. Also you have to consider the security concerns about presign URLs as the question does not mention if users are public or private.

consider the security concerns about presign URLs as the question does not mention if users are public or private.
upvoted 2 times

Yelizaveta 2 years, 4 months ago

Here it means to decouple the processes, so that the web server don't have to do the resizing, so it doesn't slow down. The customers access the web server, so the web server have to be involved in the process, and how the others already wrote, the pre-signed URL is not the right solution because, of the explanation you can read in the other comments.

And additional! "Configure the application to upload images directly from EACH USER'S BROWSER to Amazon S3 through the use of a pre-signed URL"

I am not an expert, but I can't imagine that you can store an image that an user uploads in his browser etc.
upvoted 7 times

fkie4 Highly Voted 2 years, 3 months ago

Selected Answer: BD

Why would anyone vote C? signed URL is for temporary access. also, look at the vote here: https://www.examtopics.com/discussions/amazon/view/82971-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 36 times

sheilawu 1 year, 1 month ago

I agree with you. C seems a temporary solution if a download or upload demend is urgent.
upvoted 1 times

zdi561 Most Recent 4 months, 2 weeks ago

Selected Answer: CD

see https://aws.amazon.com/blogs/compute/uploading-large-objects-to-amazon-s3-using-multipart-upload-and-transfer-acceleration/
B separate concerns and has the best performance
upvoted 1 times

salman7540 5 months, 3 weeks ago

Selected Answer: BD

BD looks accurate. Presigned URLs provide short time access to users to upload or download objects to/from S3. This approach is not feasible to renew URL every time and give it to every random users who plans to access website.
Presigned URL generally used for occasional sharing of private files.
upvoted 1 times

Tjazz04 6 months ago

Selected Answer: CD

CD is the more appropriate solution bec. when the user uploads the images, it will directly uploaded to the S3 while if BD, when the user uploads the images, it will first go to the web server then to the S3 bucket and This can cause a slow upload process since the web server is processing the download from the user, then upload to the s3 bucket.
upvoted 2 times

architect_kags 6 months, 2 weeks ago

Selected Answer: BE

While C can reduce web server load, it doesn't address image resizing, which is a critical requirement. So B & D would be the answers.
upvoted 1 times

Abdullah2004 7 months, 3 weeks ago

Selected Answer: CD

It's very clear C D
upvoted 2 times

Abdullah2004 7 months, 3 weeks ago

I'm sorry it's B D
upvoted 1 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: CD

C, D for sure
upvoted 2 times

PaulGa 9 months ago

Selected Answer: BD

Ans B, D -
1st: Upload the original images to Amazon S3
2nd: Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded to resize the image.
upvoted 3 times

bignatov 9 months, 3 weeks ago

Selected Answer: CD

I am voting for C and D.
B and D could also work, but when you upload the images to the EC2 and then to S3 it could cause additional performance and network traffic load.
upvoted 2 times

Moo 10 months, 3 weeks ago

I don't understand why the expiration time of presigned urls is being questioned. You can certainly design an upload flow that uses the SDK to create a new presigned url before uploading.
upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: BD

B and D is the way to go
upvoted 3 times

creamymangosauce 11 months ago

Selected Answer: CD

CD - No point having the instance do extra work when we can use pre signed URLs and let the user directly upload to S3, hence B is not an operationally efficient option. Furthermore B results in more traffic through the instance which is inefficient.
upvoted 2 times

jatric 11 months, 2 weeks ago

Selected Answer: BD

C is not a valid option to upload the image is to have presigned url to retrieve the file/image from S3
upvoted 1 times

MomenAWS 1 year ago

Selected Answer: BD

BD is more reasonable than CD
upvoted 3 times

soufiyane 1 year, 2 months ago

BD is the answer, idk why anyone would choose c ?? pre-signed urls are for security pursoses
upvoted 2 times

MikeJANG 1 year, 4 months ago

Selected Answer: CD

GPT4
option B would offload the storage to S3 but still involves the web server in the upload process, which does not fully address the performance issues.
upvoted 2 times

sirasdf 1 year, 3 months ago

GPT4 is wrong. It does address the performance issue with the image processing will be done by lambda and not on the server

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #111 *Topic 1*

A company recently migrated a message processing system to AWS. The system receives messages into an ActiveMQ queue running on an Amazon EC2 instance. Messages are processed by a consumer application running on Amazon EC2. The consumer application processes the messages and writes results to a MySQL database running on Amazon EC2. The company wants this application to be highly available with low operational complexity.
Which architecture offers the HIGHEST availability?

A. Add a second ActiveMQ server to another Availability Zone. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

B. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

C. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Use Amazon RDS for MySQL with Multi-AZ enabled.

D. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an Auto Scaling group for the consumer EC2 instances across two Availability Zones. Use Amazon RDS for MySQL with Multi-AZ enabled. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (98%) A

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D

Answer is D as the "HIGHEST available" and less "operational complex"
The "Amazon RDS for MySQL with Multi-AZ enabled" option excludes A and B
The "Auto Scaling group" is more available and reduces operational complexity in case of incidents (as remediation it is automated) than just adding one more instance. This excludes C.

C and D to choose from based on
D over C since is configured

upvoted 23 times

satyaammm Most Recent 5 months, 1 week ago

Selected Answer: D

Using RDS Multi-AZ is suitable for high availability and using EC2 Auto Scaling Groups is suitable for operational overhead.
upvoted 2 times

PaulGa 9 months ago

Selected Answer: D

Ans D - extends the architecture rather than complicating it: Amazon MQ with active/standby brokers configured across two Availability Zones; Auto Scaling group for the consumer EC2 instances across two Availability Zones; Amazon RDS for MySQL with Multi-AZ enabled.
upvoted 3 times

huaze_lei 9 months, 1 week ago

Selected Answer: D

- Active MQ with active/standby
- Auto scaling across 2 AZs
- RDS with Multi AZ.

These provide the highest availability and least complexity (using Amazon MQ, auto scaling and RDS, all managed services)
upvoted 3 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D is obviously the highest available.
upvoted 2 times

effiecancode 11 months, 1 week ago

Definitely it's D
upvoted 1 times

awsgeek75 1 year, 4 months ago

Selected Answer: D

D: Managed and auto-scaling, resilient and HA service for each tier. This is well-architected too.
upvoted 3 times

Ruffyit 1 year, 7 months ago

Using Amazon MQ with active/standby brokers provides highly available message queuing across AZs.

Adding an Auto Scaling group for consumer EC2 instances across 2 AZs provides highly available processing.

Using RDS MySQL with Multi-AZ provides a highly available database.

This architecture provides high availability for all components of the system - queue, processing, and database.
upvoted 3 times

prabhjot 1 year, 8 months ago

Ans is C - C. Option C uses Amazon MQ with active/standby brokers, adds an additional consumer EC2 instance, and uses Amazon RDS for MySQL with Multi-AZ enabled. Amazon RDS Multi-AZ automatically replicates your database to another AZ and provides automated failover. This ensures high availability for both the messaging system and the database. Option D- bring More scalabilty rather HA
upvoted 3 times

pentium75 1 year, 5 months ago

D automates replacement of failed instances, thus it has higher availability than C.
upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

HIGHEST availability. Definitely option D.
upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The key reasons are:

Amazon MQ active/standby brokers across AZs for queue high availability
Auto Scaling group with consumer EC2 instances across AZs for redundant processing
RDS MySQL with Multi-AZ for database high availability
This combines the HA capabilities of MQ, EC2 and RDS to maximize fault tolerance across all components. The auto scaling also provides flexibility to scale processing capacity as needed.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

D is the correct answer.

Using Amazon MQ with active/standby brokers provides highly available message queuing across AZs.

Adding an Auto Scaling group for consumer EC2 instances across 2 AZs provides highly available processing.

Using RDS MySQL with Multi-AZ provides a highly available database.

This architecture provides high availability for all components of the system - queue, processing, and database.

upvoted 3 times

james2033 1 year, 10 months ago

Selected Answer: D

Keyword Amazon RDS, has C and D. Then D has "Auto Scaling group", choose D.

upvoted 3 times

MNotABot 1 year, 11 months ago

D
With 3 options with Amazon MQ --> A is odd one out / Then ASG with M-AZ was an easy choice

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: D

Amazon MQ with active/standby brokers configured across two AZ ensures high availability for the message broker. In case of a failure in one AZ, the other AZ's broker can take over seamlessly.

Adding an ASG for the consumer EC2 instances across two AZ provides redundancy and automatic scaling based on demand. If one consumer instance becomes unavailable or if the message load increases, the ASG can automatically launch additional instances to handle the workload.

Using RDS for MySQL with Multi-AZ enabled ensures high availability for the database. Multi-AZ automatically replicates the database to a standby instance in another AZ. If a failure occurs, RDS automatically fails over to the standby instance without manual intervention.

This architecture combines high availability for the message broker (Amazon MQ), scalability and redundancy for the consumer EC2 instances (ASG), and high availability for the database (RDS Multi-AZ). It offers the highest availability with low operational complexity by leveraging managed services and automated failover mechanisms.

upvoted 3 times

Kostya 1 year, 12 months ago

Selected Answer: D

Correct answer D

upvoted 1 times

Bmarodi 2 years ago

Selected Answer: D

to achieve ha + low operational complexity, the solution architect has to choose option D, which fulfill these requirements.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #112 *Topic 1*

A company hosts a containerized web application on a fleet of on-premises servers that process incoming requests. The number of requests is growing quickly. The on-premises servers cannot handle the increased number of requests. The company wants to move the application to AWS with minimum code changes and minimum development effort.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Fargate on Amazon Elastic Container Service (Amazon ECS) to run the containerized web application with Service Auto Scaling. Use an Application Load Balancer to distribute the incoming requests. **Most Voted**

B. Use two Amazon EC2 instances to host the containerized web application. Use an Application Load Balancer to distribute the incoming requests.

C. Use AWS Lambda with a new code that uses one of the supported languages. Create multiple Lambda functions to support the load. Use Amazon API Gateway as an entry point to the Lambda functions.

D. Use a high performance computing (HPC) solution such as AWS ParallelCluster to establish an HPC cluster that can process the incoming requests at the appropriate scale.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: A

Less operational overhead means A: Fargate (no EC2), move the containers on ECS, autoscaling for growth and ALB to balance consumption.
B - requires configure EC2
C - requires add code (developpers)
D - seems like the most complex approach, like re-architecting the app to take advantage of an HPC platform.

upvoted 20 times

satyaammm Most Recent 5 months, 1 week ago

Selected Answer: A

AWS Fargate is a serverless compute engine and is fully managed by AWS. So it is more suitable here as it provides less operational overhead.

upvoted 1 times

upvoted 1 times

PaulGa 9 months ago

Selected Answer: A

And A - its containers so use FarGate, add some auto-scaling...

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A makes sense

upvoted 1 times

cosmiccliff 1 year, 7 months ago

Selected Answer: A

key = LEAST operational overhead

Fargate a serverless service fully managed by aws
https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html#:~:text=AWS%20Fargate%20is,optimize%20cluster%20packing.

upvoted 3 times

Ruffyit 1 year, 7 months ago

Less operational overhead means A: Fargate (no EC2), move the containers on ECS, autoscaling for growth and ALB to balance consumption.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

LEAST operational overhead = AWS Fargate

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

A is the best solution to meet the requirements with the least operational overhead. The key reasons are:

AWS Fargate removes the need to provision and manage servers. Fargate will automatically scale the application based on demand. This removes a significant operational burden.
Using ECS along with Fargate provides a managed orchestration layer to easily run and scale the containerized application.
The Application Load Balancer handles distribution of traffic without additional effort.
No code changes are required to move the application to Fargate. The containers can run as-is.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

AWS Fargate removes the need to provision and manage servers, allowing you to focus on deploying and running applications. Fargate will scale compute capacity up and down automatically based on application load. This removes the operational overhead of managing servers.

upvoted 1 times

james2033 1 year, 10 months ago

Selected Answer: A

Existing: "containerized web-app", "minimum code changes + minimum development effort" --> AWS Fargate + Amazon Elastic Container Services (ECS). Easy question.

upvoted 1 times

MNotABot 1 year, 11 months ago

A
Fargate, ECS, ASG, ALB....What else one will need for a nice sleep?

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: A

Option A (AWS Fargate on Amazon ECS with Service Auto Scaling) is the best choice as it provides a serverless and managed environment for your containerized web application. It requires minimal code changes, offers automatic scaling, and utilizes an Application Load Balancer for request distribution.

Option B (Amazon EC2 instances with an Application Load Balancer) requires manual management of EC2 instances, resulting in more operational overhead compared to option A.

Option C (AWS Lambda with API Gateway) may require significant code changes and restructuring, introducing complexity and potentially increasing development effort.

Option D (AWS ParallelCluster) is not suitable for a containerized web application and involves significant setup and configuration overhead.

upvoted 3 times

Jeeva28 2 years ago

Selected Answer: A

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers or clusters of Amazon EC2 instances. With Fargate, you no longer have to provision, configure, or scale clusters of virtual machines to run containers. This removes the need to choose server types, decide when to scale your clusters, or optimize cluster packing.

upvoted 1 times

studynoplay 2 years, 1 month ago

Selected Answer: A

Least Operational Overhead = Serverless

upvoted 1 times

airraid2010 2 years, 3 months ago

Selected Answer: A

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers on clusters of Amazon EC2 instances. With Fargate, you no longer have to provision, configure, or scale of virtual machines to run containers.

https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html

upvoted 1 times

Chalamalli 2 years, 4 months ago

A is correct

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: A

The best solution to meet the requirements with the least operational overhead is Option A: Use AWS Fargate on Amazon Elastic Container Service (Amazon ECS) to run the containerized web application with Service Auto Scaling. Use an Application Load Balancer to distribute the incoming requests.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #113 *Topic 1*

A company uses 50 TB of data for reporting. The company wants to move this data from on premises to AWS. A custom application in the company's data center runs a weekly data transformation job. The company plans to pause the application until the data transfer is complete and needs to begin the transfer process as soon as possible.
The data center does not have any available network bandwidth for additional workloads. A solutions architect must transfer the data and must configure the transformation job to continue to run in the AWS Cloud.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS DataSync to move the data. Create a custom transformation job by using AWS Glue.

B. Order an AWS Snowcone device to move the data. Deploy the transformation application to the device.

C. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. **Most Voted**

D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application.

Correct Answer: *C*

Community vote distribution

C (64%) D (36%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: C

A. Use AWS DataSync to move the data. Create a custom transformation job by using AWS Glue. - No BW available for DataSync, so "asap" will be weeks/months (?)
B. Order an AWS Snowcone device to move the data. Deploy the transformation application to the device. - Snowcone will just store 14TB (SSD configuration).
C. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. - SnowBall can store 80TB (ok), takes around 1 week to move the device (faster than A), and AWS Glue allows to do ETL jobs. This is the answer.
D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application. - Same as C, but the ETL job requires the deployment/configuration/maintenance of an EC2 instance, while Glue is serverless. This means D has more operational overhead than C.

upvoted 77 times

jdr75 2 years, 2 months ago

I agree. When it said "with least Operational overhead" , it does not takes in account "migration activities" neccesary to reach the "final photo/scenario". In "operational overhead" schema, you're situated in a "final scenario" and you've only take into account how do you operate it, and if the operation of that scheme is ALIGHTED (least effort to operate than original scenario), that's the desired state.

upvoted 6 times

remand 2 years, 4 months ago

I disagree on D. transformation job is already in place.so, all you have to do is deploy and run on ec2.
C takes more effort to build Glue process, like reinventing the wheel . this is unnecessary

upvoted 9 times

xxichlas 1 year, 1 month ago

the differene between snowball with ec2 and without ec2 is $200 (for us east 1). fair to assume aws glue will not be more than that

upvoted 2 times

goodmail Highly Voted 2 years, 5 months ago

Selected Answer: D

Why C? This answer misses the part between SnowBall and AWS Glue.
D at least provides a full-step solution that copies data in snowball device, and installs the custom application in device's EC2 to do the transformation job.

upvoted 17 times

happpieee 1 year, 3 months ago

AWS Glue is not part of SnowBall Edge AWS services it can run within. Check it out here :
https://docs.aws.amazon.com/snowball/latest/developer-guide/whatisedge.html

upvoted 3 times

pentium75 1 year, 5 months ago

Because AWS Glue means less "operational (!) overhead" than running an EC2 instance.

upvoted 6 times

c12ab95 Most Recent 4 weeks ago

Selected Answer: D

Option Shortcoming
A AWS DataSync requires network bandwidth, which is unavailable.
B Snowcone’s maximum capacity (14 TB SSD) is insufficient for 50 TB.
C AWS Glue would require rewriting the custom application, increasing operational overhead.
Option D minimizes operational overhead by leveraging Snowball for offline transfer and EC2 for running the existing application, ensuring a seamless transition to AWS.

upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: C

One requirement is to run transformation with the data in aws, that is the job of glue

upvoted 1 times

architect_kags 6 months, 2 weeks ago

Selected Answer: D

Since the transformation application already there, Snowball Edge devices with compute capability allow to run applications directly on the device. This means the transformation job can start locally on the Snowball Edge device if needed or directly after the data transfer to AWS, minimizing delays. AWS Glue for transformation adds additional refactoring work and operational overhead.

upvoted 3 times

tom_cruise 6 months, 2 weeks ago

Selected Answer: D

The only way C can work is by using this: Amazon S3 adapter — Use for programmatic data transfer in to and out of AWS using the Amazon S3 API for Snowball Edge, but it is not indicated in the answer. If it includes the step to transfer the data from the Snowball to S3, then C would be a better answer.
upvoted 2 times

0de7d1b 6 months, 3 weeks ago

Selected Answer: D

Application still need EC2 instance along data transfer with Snowball
upvoted 2 times

PaulGa 9 months ago

Selected Answer: D

Ans D - but I suspect it should be Ans C because, as stated by 123jhlo (1yr 11mth ago),
D is not serverless:

"**C**. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. - SnowBall can store 80TB (ok), takes around 1 week to move the device (faster than A), and AWS Glue allows to do ETL jobs."
D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application. - Same as C, but the ETL job requires the deployment/configuration/maintenance of an EC2 instance, while Glue is serverless. This means D has more operational overhead than C."
upvoted 3 times

bignatov 9 months, 3 weeks ago

Selected Answer: D

A and D are wrong. D is the correct answer.
Why not C:

AWS Snowball Edge is a physical device designed for transferring large amounts of data to and from AWS. It includes some compute capabilities, such as running AWS Lambda functions, AWS IoT Greengrass, and EC2 instances, but it does not support AWS Glue.

AWS Glue is a fully managed ETL (Extract, Transform, Load) service that runs within the AWS Cloud. It is designed to work with data stored in AWS services like Amazon S3, Amazon RDS, and Amazon Redshift, among others. AWS Glue is not available on edge devices like Snowball Edge.
upvoted 3 times

Lexxo 3 months, 3 weeks ago

Yes, AWS Glue can connect to AWS Snowball Edge. AWS Snowball Edge is a data transfer and edge computing device that allows you to move large amounts of data into and out of AWS. You can use AWS Glue to create ETL jobs that read data from Snowball Edge and write it to other AWS services like Amazon S3, Amazon Redshift, or Amazon RDS. --- FROM Copilot
upvoted 1 times

bignatov 9 months, 3 weeks ago

just correction A and B are wrong and D is the correct answer.
upvoted 1 times

lofzee 1 year ago

gotta be C surely..... LEAST operational overhead.
EC2 = operational overhead.
AWS Glue = managed service with transformation capabilities.
upvoted 2 times

f761d0e 1 year, 1 month ago

The q states that the custom job must remain, so glue is out. Seems like D is the only option as DataSync needs bandwidth.
upvoted 4 times

vip2 1 year, 3 months ago

Selected Answer: C

Using an EC2 instance instead of a managed service like AWS Glue will include more operational overhead for the organization.

organization.

upvoted 2 times

Femmyte 1 year, 4 months ago

Selected Answer: D

The answer is D because of the following key points
1. A custom application in the company's data center runs a weekly data transformation job. Which means that the company already has an application that runs the transformation.
2. A solutions architect must transfer the data and must configure the transformation job to continue to run in the AWS Cloud. This shows that the only responsibility of the architect is to transfer the data and configure the existing application to run on the EC2 the architect is going to deploy.

upvoted 3 times

awsgeek75 1 year, 4 months ago

Selected Answer: C

A: Cannot be done because no bandwidth
B: Snowcone is probably to small
D: Doable by EC2 is overhead for transformation when Glue is an option

C: Is correct as Snowball Edge Storage Optimised device is good for storage and Glue can transform once the data is available

upvoted 2 times

bujuman 1 year, 5 months ago

Selected Answer: C

C best suit for:
ETL jobs with LEAST operational overhead.
For my understanding, we need here to avoir operation or maintenance burden of the solution

upvoted 1 times

Shalen 1 year, 6 months ago

Selected Answer: D

we use snowball to copy 50 PB
"The company plans to pause the application until the data transfer is complete "
and least over head " hence C would be reinventing the weel

upvoted 1 times

slimen 1 year, 7 months ago

Selected Answer: D

which is faster?
- setup a glue cluster and adapt it to do the same analytical stuff as the original app
- simpley run the same app in an EC2 instance?

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #114 *Topic 1*

A company has created an image analysis application in which users can upload photos and add photo frames to their images. The users upload images and metadata to indicate which photo frames they want to add to their images. The application uses a single Amazon EC2 instance and Amazon DynamoDB to store the metadata.
The application is becoming more popular, and the number of users is increasing. The company expects the number of concurrent users to vary significantly depending on the time of day and day of week. The company must ensure that the application can scale to meet the needs of the growing user base.
Which solution meats these requirements?

A. Use AWS Lambda to process the photos. Store the photos and metadata in DynamoDB.

B. Use Amazon Kinesis Data Firehose to process the photos and to store the photos and metadata.

C. Use AWS Lambda to process the photos. Store the photos in Amazon S3. Retain DynamoDB to store the metadata.
Most Voted

D. Increase the number of EC2 instances to three. Use Provisioned IOPS SSD (io2) Amazon Elastic Block Store (Amazon EBS) volumes to store the photos and metadata.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

MXB05 Highly Voted 2 years, 8 months ago

Selected Answer: C

Do not store images in databases ;)... correct answer should be C

upvoted 48 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: C

Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

Option A does not provide an appropriate solution for storing the photos, as DynamoDB is not suitable for storing large binary data like images.

Option B is more focused on real-time streaming data processing and is not the ideal service for processing and storing photos and metadata in this use case.

Option D involves manual scaling and management of EC2 instances, which is less flexible and more labor-intensive compared to the serverless nature of Lambda. It may not efficiently handle the varying number of concurrent users and can introduce higher operational overhead.

In conclusion, option C provides the best solution for scaling the application to meet the needs of the growing user base by leveraging the scalability and durability of Lambda, S3, and DynamoDB.

upvoted 17 times

michaelmorar Most Recent 5 months, 2 weeks ago

Selected Answer: C

Bad question - it does not specify a requirement for storing the photos nor does it specify where the old application does so. C is the best answer absent further requirements.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: C

Ans C - as well explained by cookieMr (1yr 2mth ago):
"Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

Option A does not provide an appropriate solution for storing the photos, as DynamoDB is not suitable for storing large binary data like images.

...option C provides the best solution for scaling the application to meet the needs of the growing user base by leveraging the scalability and durability of Lambda, S3, and DynamoDB."

upvoted 2 times

pranavff_examtopics_1993 9 months ago

Selected Answer: C

DynamoDB should not be used for storing images and files in general. Answer should be C.

upvoted 2 times

zliang14 9 months, 1 week ago

DynamoDB is not designed for storing large objects like photos. Amazon S3 is the correct storage service for photos.

upvoted 2 times

snk27 10 months, 3 weeks ago

Selected Answer: C

s3 bucket is good option to store images

upvoted 3 times

Gape4 12 months ago

Selected Answer: C

C is correct answer.

upvoted 1 times

f761d0e 1 year, 1 month ago

C. DB is for data, not for photos. Kinesis doesn't store, it processes streaming.

upvoted 2 times

MehulKapadia 1 year, 2 months ago

Selected Answer: C

DynamoDB items max size is 400kb. so A cannot be right answer.

Correct Answer is C

upvoted 2 times

farnamjam 1 year, 5 months ago

Selected Answer: C

Max size for DDB entry is 400KB.

upvoted 2 times

aptx4869 1 year, 7 months ago

Selected Answer: C

Images (Object) should go in S3 and metadata should go in database (DynamoDB)

upvoted 2 times

Ruffyit 1 year, 7 months ago

Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

upvoted 2 times

Ferna 1 year, 7 months ago

Selected Answer: C

Solution C

upvoted 1 times

David_Ang 1 year, 8 months ago

Selected Answer: C

i think is only a confusion of the admin, because it has more sense to store the photos in a S3 bucket is logic.

upvoted 1 times

tom_cruise 1 year, 8 months ago

Selected Answer: C

A does not store data.

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

I stopped at option C

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #115 *Topic 1*

A medical records company is hosting an application on Amazon EC2 instances. The application processes customer data files that are stored on Amazon S3. The EC2 instances are hosted in public subnets. The EC2 instances access Amazon S3 over the internet, but they do not require any other network access.
A new requirement mandates that the network traffic for file transfers take a private route and not be sent over the internet.
Which change to the network architecture should a solutions architect recommend to meet this requirement?

A. Create a NAT gateway. Configure the route table for the public subnets to send traffic to Amazon S3 through the NAT gateway.

B. Configure the security group for the EC2 instances to restrict outbound traffic so that only traffic to the S3 prefix list is permitted.

C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets. **Most Voted**

D. Remove the internet gateway from the VPC. Set up an AWS Direct Connect connection, and route traffic to Amazon S3 over the Direct Connect connection.

Correct Answer: *C*

Community vote distribution

C (95%) D (5%)

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: C

Option A (creating a NAT gateway) would not meet the requirement since it still involves sending traffic to S3 over the internet. NAT gateway is used for outbound internet connectivity from private subnets, but it doesn't provide a private route for accessing S3.

Option B (configuring security groups) focuses on controlling outbound traffic using security groups. While it can restrict outbound traffic, it doesn't provide a private route for accessing S3.

Option D (setting up Direct Connect) involves establishing a dedicated private network connection between the on-premises environment and AWS. While it offers private connectivity, it is more suitable for hybrid scenarios and not necessary for achieving private access to S3 within the VPC.

In summary, option C provides a straightforward solution by moving the EC2 instances to private subnets, creating a VPC endpoint for S3, and linking the endpoint to the route table for private subnets. This ensures that file transfer traffic between the EC2 instances and S3 remains within the private network without going over the internet.

upvoted 16 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: C

The correct answer is C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.

To meet the new requirement of transferring files over a private route, the EC2 instances should be moved to private subnets, which do not have direct access to the internet. This ensures that the traffic for file transfers does not go over the internet.

To enable the EC2 instances to access Amazon S3, a VPC endpoint for Amazon S3 can be created. VPC endpoints allow resources within a VPC to communicate with resources in other services without the traffic being sent over the internet. By linking the VPC endpoint to the route table for the private subnets, the EC2 instances can access Amazon S3 over a private connection within the VPC.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A (Create a NAT gateway) would not work, as a NAT gateway is used to allow resources in private subnets to access the internet, while the requirement is to prevent traffic from going over the internet.

Option B (Configure the security group for the EC2 instances to restrict outbound traffic) would not achieve the goal of routing traffic over a private connection, as the traffic would still be sent over the internet.

Option D (Remove the internet gateway from the VPC and set up an AWS Direct Connect connection) would not be necessary, as the requirement can be met by simply creating a VPC endpoint for Amazon S3 and routing traffic through it.

upvoted 1 times

Kayamables 2 years, 5 months ago

How about the question of moving the instances across subnets. Because according to AWS you can't do it.
https://aws.amazon.com/premiumsupport/knowledge-center/move-ec2-instance/#:~:text=It%27s%20not%20possible%20to%20move,%2C%20Availability%20Zone%2C%20or%20VPC.
Kindly clarify. Maybe I miss something.

upvoted 2 times

pentium75 1 year, 5 months ago

You can't just change the subnet in instance settings, but this article mentions how you CAN move the instance manually.

upvoted 2 times

Vandaman Most Recent 3 months, 3 weeks ago

Selected Answer: C

C is the best Answer. The requirement is for file transfers to take a private route and not be sent over the internet. This eliminates A and B. D is for Hybrid solutions.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: D

Ans C - I was going for Ans D...
...but as well explained by Buruguduystunstugudunstuy (1 year, 8 mth ago), C is simpler:
"Option C: Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.
To meet the new requirement of transferring files over a private route, the EC2 instances should be moved to private subnets, which do not have direct access to the internet. This ensures that the traffic for file transfers does not go over the internet.
"Option D (Remove the internet gateway from the VPC and set up an AWS Direct Connect connection) would not be necessary, as the requirement can be met by simply creating a VPC endpoint for Amazon S3 and routing traffic through it."

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: C

C is the correct answer.

upvoted 1 times

Ruffyit 1 year, 7 months ago

C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.
upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.
upvoted 1 times

sand444 1 year, 9 months ago

Selected Answer: C

link VPC endpoint in route tables ---- EC2 instance to communicate S3 with a private connection in VPC
upvoted 2 times

DavidNamy 2 years, 5 months ago

Selected Answer: C

According to the well-designed framework, option C is the safest and most efficient option.
upvoted 3 times

career360guru 2 years, 5 months ago

Selected Answer: C

Option C
upvoted 1 times

ocbn3wby 2 years, 6 months ago

C is correct.
There is no requirement for public access from internet.

Application must be moved in Private subnet. This is a prerequisite in using VPC endpoints with S3
https://aws.amazon.com/blogs/storage/managing-amazon-s3-access-with-vpc-endpoints-and-s3-access-points/
upvoted 5 times

Wpcorgan 2 years, 6 months ago

C is correct
upvoted 1 times

Jtic 2 years, 7 months ago

Selected Answer: C

Use VPC endpoint
upvoted 1 times

Jtic 2 years, 7 months ago

Selected Answer: C

User VPC endpoint and make the EC2 private
upvoted 1 times

Jtic 2 years, 7 months ago

Use VPC endpoint
upvoted 1 times

backbencher2022 2 years, 7 months ago

Selected Answer: C

VPC endpoint is the best choice to route S3 traffic without traversing internet. Option A alone can't be used as NAT Gateway requires an internet gateway for outbound internet traffic. Option B would still require traversing through internet and option D is also not a suitable solution

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #116 *Topic 1*

A company uses a popular content management system (CMS) for its corporate website. However, the required patching and maintenance are burdensome. The company is redesigning its website and wants anew solution. The website will be updated four times a year and does not need to have any dynamic content available. The solution must provide high scalability and enhanced security.
Which combination of changes will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Configure Amazon CloudFront in front of the website to use HTTPS functionality. **Most Voted**

B. Deploy an AWS WAF web ACL in front of the website to provide HTTPS functionality.

C. Create and deploy an AWS Lambda function to manage and serve the website content.

D. Create the new website and an Amazon S3 bucket. Deploy the website on the S3 bucket with static website hosting enabled. **Most Voted**

E. Create the new website. Deploy the website by using an Auto Scaling group of Amazon EC2 instances behind an Application Load Balancer.

Correct Answer: *AD*

Community vote distribution

AD (89%) Other (11%)

Comments

palermo777 Highly Voted 2 years, 7 months ago

A -> We can configure CloudFront to require HTTPS from clients (enhanced security)
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/using-https-viewers-to-cloudfront.html
D -> storing static website on S3 provides scalability and less operational overhead, then configuration of Application LB and EC2 instances (hence E is out)

B is out since AWS WAF Web ACL does not to provide HTTPS functionality, but to protect HTTPS only.

upvoted 36 times

Six_Fingered_Jose Highly Voted 2 years, 7 months ago

Selected Answer: AD

agree with A and D

static website -> obviously S3, and S3 is super scalable

static website -> obviously S3, and S3 is super scalable
CDN -> CloudFront obviously as well, and with HTTPS security is enhanced.

B does not make sense because you are not replacing the CDN with anything,
E works too but takes too much effort and compared to S3, S3 still wins in term of scalability. plus why use EC2 when you are only hosting static website

upvoted 10 times

Lalo 2 years ago

Amazon CloudFront is for Securely deliver content with low latency and high transfer speeds
But what about the SQLinjection XSS attacks? we use WAF and olso use HTTPS
https://www.f5.com/glossary/web-application-firewall-waf#:~:text=A%20WAF%20protects%20your%20web,and%20what%20traffic%20is%20safe.
WAF protects your web apps by filtering, monitoring, and blocking any malicious HTTP/S traffic traveling to the web application, and prevents any unauthorized data from leaving the app.
Answer is WAF Not Cloudfront

upvoted 3 times

aussiehoa 2 years, 1 month ago

does not need to have any dynamic content available

upvoted 2 times

jaradat02 Most Recent 10 months, 3 weeks ago

Selected Answer: AD

A and D is the safest combination.

upvoted 2 times

David_Ang 1 year, 8 months ago

Selected Answer: AD

these answers are the most common use case for real companies, is like the answers that have more sense

upvoted 3 times

tom_cruise 1 year, 8 months ago

Selected Answer: AD

Web Application Firewall creates rules to block attacks, but it does not create HTTPS. It can only allow HTTPS inbound traffic.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: AD

Scalability, enhanced security and less operational overhead = CloudFront with HTTPS
Scalability and less operational overhead = S3 bucket with static website hosting

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: AD

A. Amazon CloudFront provides scalable content delivery with HTTPS functionality, meeting security and scalability requirements.

D. Deploying the website on an Amazon S3 bucket with static website hosting reduces operational overhead by eliminating server maintenance and patching.

Why other options are incorrect:

B. AWS WAF does not provide HTTPS functionality or address patching and maintenance.

C. Using AWS Lambda introduces complexity and does not directly address patching and maintenance.

E. Managing EC2 instances and an Application Load Balancer increases operational overhead and does not minimize patching and maintenance tasks.

In summary, configuring Amazon CloudFront for HTTPS and deploying on Amazon S3 with static website hosting provide security, scalability, and reduced operational overhead.

upvoted 2 times

upvoted 2 times

beginnercloud 2 years ago

Selected Answer: AD

AD

A for enhanced security D for static content

upvoted 2 times

studynoplay 2 years, 1 month ago

Selected Answer: AD

LEAST operational overhead = Serverless
https://aws.amazon.com/serverless/

upvoted 4 times

angolateoria 2 years, 1 month ago

AD misses the operational part, how can the app work without a lambda function, an EC2 instance or something?

upvoted 1 times

darn 2 years, 1 month ago

Selected Answer: AD

people do not seem to get the LEAST OPERATIONAL OVERHEAD statement, many people keep voting for options that bring far too Op work

upvoted 2 times

channn 2 years, 2 months ago

Selected Answer: AD

A for enhanced security
D for static content

upvoted 3 times

Erbug 2 years, 2 months ago

Since Amazon S3 is unlimited and you pay as you go so it means there will be no limit to scale as long as your data is going to grow, so D is one of the correct answers and another correct answer is A, because of this:
https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteHosting.html

so my answer is AD.

upvoted 2 times

ManOnTheMoon 2 years, 3 months ago

I vote A & C for the reason being least operational overhead.

upvoted 1 times

Yelizaveta 2 years, 4 months ago

Selected Answer: AD

Here a perfect explanation:
https://aws.amazon.com/premiumsupport/knowledge-center/cloudfront-serve-static-website/

upvoted 4 times

Abdel42 2 years, 4 months ago

Selected Answer: AD

Simple and secure

upvoted 1 times

remand 2 years, 4 months ago

Selected Answer: AD

D. Create the new website and an Amazon S3 bucket. Deploy the website on the S3 bucket with static website hosting enabled.
A. Configure Amazon CloudFront in front of the website to use HTTPS functionality.

By deploying the website on an S3 bucket with static website hosting enabled, the company can take advantage of the high scalability and cost-efficiency of S3 while also reducing the operational overhead of managing and patching a CMS.
By configuring Amazon CloudFront in front of the website, it will automatically handle the HTTPS functionality, this way the company can have a secure website with very low operational overhead.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #117 *Topic 1*

A company stores its application logs in an Amazon CloudWatch Logs log group. A new policy requires the company to store all application logs in Amazon OpenSearch Service (Amazon Elasticsearch Service) in near-real time.
Which solution will meet this requirement with the LEAST operational overhead?

A. Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service). **Most Voted**

B. Create an AWS Lambda function. Use the log group to invoke the function to write the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

C. Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery streams sources. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

D. Install and configure Amazon Kinesis Agent on each application server to deliver the logs to Amazon Kinesis Data Streams. Configure Kinesis Data Streams to deliver the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

Correct Answer: *A*

Community vote distribution

A (67%) C (31%) Oth

Comments

Six_Fingered_Jose Highly Voted 2 years, 7 months ago

Selected Answer: A

answer is A
https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

> You can configure a CloudWatch Logs log group to stream data it receives to your Amazon OpenSearch Service cluster in NEAR REAL-TIME through a CloudWatch Logs subscription

least overhead compared to kinesis
upvoted 100 times

Vandaman 3 months, 3 weeks ago

Thank you for the link - clear answer
upvoted 1 times

upvoted 1 times

lofzee 1 year ago

good enough for me

upvoted 2 times

HayLLlHuK 2 years, 5 months ago

Zerotn3 is right! There should be a Lambda for writing into ES

upvoted 1 times

UWSFish 2 years, 7 months ago

Great link. Convinced me

upvoted 5 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: C

The correct answer is C: Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery stream source. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

This solution uses Amazon Kinesis Data Firehose, which is a fully managed service for streaming data to Amazon OpenSearch Service (Amazon Elasticsearch Service) and other destinations. You can configure the log group as the source of the delivery stream and Amazon OpenSearch Service as the destination. This solution requires minimal operational overhead, as Kinesis Data Firehose automatically scales and handles data delivery, transformation, and indexing.

upvoted 19 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A: Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to set up and manage the subscription and ensure that the logs are delivered in near-real time.

Option B: Create an AWS Lambda function. Use the log group to invoke the function to write the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to set up and manage the Lambda function and ensure that it scales to handle the incoming logs.

Option D: Install and configure Amazon Kinesis Agent on each application server to deliver the logs to Amazon Kinesis Data Streams. Configure Kinesis Data Streams to deliver the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to install and configure the Kinesis Agent on each application server and set up and manage the Kinesis Data Streams.

upvoted 3 times

ocbn3wby 2 years, 4 months ago

https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

upvoted 2 times

Lalo 2 years ago

ANSWER A
https://docs.aws.amazon.com/opensearch-service/latest/developerguide/integrations.html
You can use CloudWatch or Kinesis, but in the Kinesis description it never says real time, however in the Cloudwatch description it does say Real time ""You can load streaming data from CloudWatch Logs to your OpenSearch Service domain by using a CloudWatch Logs subscription . For information about Amazon CloudWatch subscriptions, see Real-time processing of log data with subscriptions.""

upvoted 4 times

Mischi Most Recent 5 months, 3 weeks ago

Selected Answer: D

In summary, the CloudWatch Logs → Kinesis Data Firehose → Amazon OpenSearch Service (option C) integration is the path recommended by AWS for this type of case. It allows for near real-time transmission, automatic scaling and relatively simple configuration, with the lowest operational overhead .

upvoted 1 times

0de7d1b 6 months, 3 weeks ago

Selected Answer: A

CloudWatch Logs subscription filter: This is the most straightforward way to stream logs from a CloudWatch Logs group to Amazon OpenSearch Service (Amazon Elasticsearch Service) in near real-time. It eliminates the need for additional components or complex configurations, reducing operational overhead.
Direct integration: CloudWatch Logs can directly stream logs to OpenSearch Service without requiring intermediate services, making it a simple and efficient solution.
Low operational overhead: Once set up, the subscription filter automatically forwards logs to OpenSearch Service with minimal maintenance.
upvoted 2 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: C

C for sure
Simplicity: Kinesis Data Firehose is a managed service that handles the task of capturing, transforming, and loading data into destinations like Amazon OpenSearch Service. This eliminates the need for complex configuration and management.
Scalability: Kinesis Data Firehose can automatically scale to handle varying data volumes, ensuring that logs are ingested in near-real time.
Cost-effectiveness: Kinesis Data Firehose is a pay-as-you-go service, making it a cost-effective option for log ingestion and analysis.
upvoted 2 times

tonybuivannghia 8 months, 3 weeks ago

Selected Answer: C

I think C is correct because the cloud watch subscription can't stream directly to OpenSearch, it is via Lambda, SNS, FireHouse,....
upvoted 3 times

Tieri 8 months, 3 weeks ago

You can configure a log group in Amazon CloudWatch Logs, so you can stream data to your Amazon OpenSearch Service cluster in near real-time.
upvoted 1 times

Johnoppong101 9 months, 2 weeks ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html
upvoted 2 times

KTEgghead 10 months, 2 weeks ago

Selected Answer: A

Configure a CloudWatch Logs log group to stream data directly to the Amazon OpenSearch Service cluster. This can be done through a CloudWatch Logs subscription, which allows for real-time processing of log data.
upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the correct answer, CloudWatch offers a subscription where you can stream data to other AWS services
upvoted 1 times

Seb888 11 months ago

Selected Answer: C

Correct Answer:
C. Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery stream's source. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

Explanation:
Amazon Kinesis Data Firehose is a fully managed service for delivering real-time streaming data to destinations such as Amazon OpenSearch Service. It requires minimal setup and management, making it a low-overhead solution.
By configuring the log group as the source for the Kinesis Data Firehose delivery stream and Amazon OpenSearch Service as the destination, logs can be delivered in near-real time with built-in reliability and scalability.
upvoted 2 times

jatric 11 months, 2 weeks ago

Selected Answer: A

easy enough to figure out. Option A
upvoted 1 times

ChymKuBoy 11 months, 4 weeks ago

Selected Answer: A

A for sure
upvoted 1 times

824c449 1 year, 1 month ago

Selected Answer: C

It can natively connect to CloudWatch Logs as a source and OpenSearch Service as a destination, handling the delivery of logs efficiently and with minimal setup. This approach offers the least operational overhead by simplifying the data transfer pipeline with automatic scaling and error handling.
upvoted 1 times

zinabu 1 year, 1 month ago

Selected Answer: A

You can configure a CloudWatch Logs log group to stream data it receives to your Amazon OpenSearch Service cluster in near real-time through a CloudWatch Logs subscription.

here is the link/; https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html
upvoted 1 times

OctavioBatera 1 year, 2 months ago

Selected Answer: A

Answer A.
This doc clarifies the subject: https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html
upvoted 1 times

CloudLearner01 1 year, 3 months ago

A is correct
You can configure a CloudWatch Logs log group to stream data it receives to your Amazon OpenSearch Service cluster in near real-time through a CloudWatch Logs subscription.

https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html
upvoted 1 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #118 *Topic 1*

A company is building a web-based application running on Amazon EC2 instances in multiple Availability Zones. The web application will provide access to a repository of text documents totaling about 900 TB in size. The company anticipates that the web application will experience periods of high demand. A solutions architect must ensure that the storage component for the text documents can scale to meet the demand of the application at all times. The company is concerned about the overall cost of the solution.
Which storage solution meets these requirements MOST cost-effectively?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Elastic File System (Amazon EFS)

C. Amazon OpenSearch Service (Amazon Elasticsearch Service)

D. Amazon S3 **Most Voted**

Correct Answer: *D*

Community vote distribution

D (98%) C

Comments

Azure55 Highly Voted 1 year, 7 months ago

Selected Answer: D

the cost of S3<EFS<EBS

upvoted 16 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: D

Amazon S3 (Simple Storage Service) is a highly scalable and cost-effective storage service. It is well-suited for storing large amounts of data, such as the 900 TB of text documents mentioned in the scenario. S3 provides high durability, availability, and performance.

Option A (Amazon EBS) is block storage designed for individual EC2 instances and may not scale as seamlessly and cost-effectively as S3 for large amounts of data.

Option B (Amazon EFS) is a scalable file storage service, but it may not be the most cost-effective option compared to S3, especially for the anticipated storage size of 900 TB.

Option C (Amazon OpenSearch Service) is a search and analytics service and may not be suitable as the primary storage solution for the text documents.

In summary, Amazon S3 is the recommended choice as it offers high scalability, cost-effectiveness, and durability for storing the large repository of text documents required by the web application.

upvoted 11 times

PaulGa Most Recent 9 months ago

Selected Answer: D

Ans D - Amazon S3 is highly scalable, cost-effective storage service, well-suited for large amounts of data. It is highly durable, highly available, and offers good performance.
By comparison, EFS (option B) could do it but is more expensive...

upvoted 3 times

Americanman 9 months, 4 weeks ago

S3 can be a good option for storing text documents. It allows users to store any file type as objects. (documents, videos, images)

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

Using EFS would obviously be the optimal case, we have to use s3 to fulfill the cost efficiency requirement.

upvoted 3 times

theochan 1 year, 5 months ago

Option A : EBS can't be multi-AZ
Option B: EFS is expensive
Option C: ElasticSearch is not for storing

upvoted 4 times

awashenko 1 year, 8 months ago

Selected Answer: D

D is the only real solution here. S3 is the cheapest option for storage and it can scale indefinitely.

upvoted 5 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

MOST cost-effective = S3 (unless explicitly stated in the requirements)

upvoted 4 times

Jeeva28 2 years ago

Selected Answer: D

900 in the question to divert our Thinking.When you have keyword least in question S3 will be only thing we should look

upvoted 3 times

Abrar2022 2 years ago

EFS and S3 meet the requirements but S3 is a better option because it is cheaper.

upvoted 2 times

studynoplay 2 years, 1 month ago

Selected Answer: D

MOST cost-effective = S3 (unless explicitly stated in the requirements)

upvoted 3 times

Robrobtutu 2 years, 1 month ago

Selected Answer: D

S3 is the cheapest and most scalable.

upvoted 2 times

jdr75 2 years, 2 months ago

Selected Answer: C

Now in OpenSearch you can reach at 3 PB so option C is better.
With S3 in an intensive scenario the costs of retriving the buckets could be high.
Yes OpenSearch is NOT cheap but this has to be analysed carefully.
So, I opt "C" to increase the discussion.

With UltraWarm, you can retain up to 3 PB of data on a single Amazon OpenSearch Service cluster, while reducing your cost per GB by nearly 90% compared to the warm storage tier. You can also easily query and visualize the data in your Kibana interface (version 7.10 and earlier) or OpenSearch Dashboards. Analyze both your recent (weeks) and historical (months or years) log data without spending hours or days restoring archived logs.

https://aws.amazon.com/es/opensearch-service/features/

upvoted 2 times

Dr_Chomp 2 years, 2 months ago

EFS is a good option but expensive alongside S3 and customer concerned about cost - thus: S3 (D)

upvoted 3 times

frenzoid 2 years, 2 months ago

I wonder why people choose S3, yet S3 max capacity is 5TB .

upvoted 3 times

frenzoid 2 years, 2 months ago

My bad, the 5TB limit is for individual files. S3 has virtually unlimited storage capacity.

upvoted 8 times

Help2023 2 years, 3 months ago

Selected Answer: D

A. It is Not a block storage
B. It is Not a file storage
C. Opensearch is useful but can only accommodate up to 600TiB and is mainly for search and anaytics.
D. S3 is more cost effective than all and can handle all objects like Block, File or Text.

upvoted 5 times

remand 2 years, 4 months ago

Selected Answer: D

D. Amazon S3

Amazon S3 is an object storage service that can store and retrieve large amounts of data at any time, from anywhere on the web. It is designed for high durability, scalability, and cost-effectiveness, making it a suitable choice for storing a large repository of text documents. With S3, you can store and retrieve any amount of data, at any time, from anywhere on the web, and you can scale your storage up or down as needed, which will help to meet the demand of the web application. Additionally, S3 allows you to choose between different storage classes, such as standard, infrequent access, and archive, which will enable you to optimize costs based on your specific use case.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #119 *Topic 1*

A global company is using Amazon API Gateway to design REST APIs for its loyalty club users in the us-east-1 Region and the ap-southeast-2 Region. A solutions architect must design a solution to protect these API Gateway managed REST APIs across multiple accounts from SQL injection and cross-site scripting attacks.
Which solution will meet these requirements with the LEAST amount of administrative effort?

A. Set up AWS WAF in both Regions. Associate Regional web ACLs with an API stage.

B. Set up AWS Firewall Manager in both Regions. Centrally configure AWS WAF rules. **Most Voted**

C. Set up AWS Shield in bath Regions. Associate Regional web ACLs with an API stage.

D. Set up AWS Shield in one of the Regions. Associate Regional web ACLs with an API stage.

Correct Answer: *B*

Community vote distribution

B (69%) A (30%) C

Comments

Gil80 Highly Voted 2 years, 7 months ago
Selected Answer: B

If you want to use AWS WAF across accounts, accelerate WAF configuration, automate the protection of new resources, use Firewall Manager with AWS WAF
upvoted 43 times

slimen 1 year, 7 months ago

they didn't mention multiple accounts! only 2 regoins
upvoted 3 times

lofzee 1 year ago

wtf? the question says
"to protect these API Gateway managed REST APIs across multiple accounts from SQL injection and cross-site scripting attack"
upvoted 9 times

baku98 1 year, 6 months ago

B is wrong: AWS Firewall Manager cannot create security policies across regions.
Q: Can I create security policies across regions? No, AWS Firewall Manager security policies are region specific. Each Firewall Manager policy can only include resources available in that specified AWS Region. You can create a new policy for each region where you operate.
https://aws.amazon.com/firewall-manager/faqs/#:~:text=No%2C%20AWS%20Firewall%20Manager%20security,in%20that%20specified%20AWS%20Region.
upvoted 8 times

mauroicardi 1 year, 3 months ago

AWS Firewall Manager is integrated with AWS Organizations so you can enable AWS WAF rules, AWS Shield Advanced protections, VPC security groups, AWS Network Firewalls, and Amazon Route 53 Resolver DNS Firewall rules across multiple AWS accounts and resources from a single place.
upvoted 4 times

pentium75 1 year, 5 months ago

That's why B says that you "set up AWS Firewall Manager IN BOTH REGIONS". Still you can "centrally configure" WAF per region, so that you don't have to attach WAF to every individual API.
upvoted 5 times

Nigma Highly Voted 2 years, 7 months ago

B

Using AWS WAF has several benefits. Additional protection against web attacks using criteria that you specify. You can define criteria using characteristics of web requests such as the following:
Presence of SQL code that is likely to be malicious (known as SQL injection).
Presence of a script that is likely to be malicious (known as cross-site scripting).

AWS Firewall Manager simplifies your administration and maintenance tasks across multiple accounts and resources for a variety of protections.

https://docs.aws.amazon.com/waf/latest/developerguide/what-is-aws-waf.html
upvoted 18 times

JayBee65 2 years, 5 months ago

Q: Can I create security policies across regions?

No, AWS Firewall Manager security policies are region specific. Each Firewall Manager policy can only include resources available in that specified AWS Region. You can create a new policy for each region where you operate.

So you could not centrally (i.e. in one place) configure policies, you would need to do this is each region
upvoted 4 times

pentium75 1 year, 5 months ago

"Centrally" on the Firewall Manager per region, as opposed to individually for every single API.
upvoted 2 times

AwsAbhiKumar Most Recent 3 months, 3 weeks ago

Selected Answer: A

AWF has Web ACL which has a rule regarding HTTP header, HHTP body or URL string protects from common attack - SQL Injection and cross site scripting.
upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: A

C&D are out of question as AWS Shield is only for DDoS attack prevention.
B is not supported - one single policy cannot be used for multiple regions.
That leaves out A.
upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

OK I stand corrected: the choice does say "Centrally manager rules" and not "use a sigle policy". In that view B is indeed the

most appropriate.
upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

This is a tricky one. Again. AWS WAF is a product specifically for preventing SQL Injection attacks. So maybe A is indeed the right choice. I believe A is indeed the right answer now, after referring to the doc.
https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-sqli-match.html
upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: B

Centralized control with Firewall Manager means you can create and manage WAF rules once and apply them across multiple accounts and Regions, ensuring consistency and compliance.
Firewall Manager automatically applies policies to new and existing resources across accounts, reducing the effort of manually associating web ACLs in each Region and account.
upvoted 1 times

FlyingHawk 4 months, 3 weeks ago

Selected Answer: B

If only a few accounts, A is more straight. but the question mentions "multiple accounts", may be B is better choice.
upvoted 1 times

tom_cruise 6 months, 2 weeks ago

Selected Answer: C

The keywords here is "across multiple accounts", not "across multiple regions".
upvoted 1 times

0de7d1b 6 months, 3 weeks ago

Selected Answer: B

AWS Firewall Manager:

Centralized Management: Allows you to centrally manage security policies across multiple accounts and regions.
WAF Rule Configuration: Enables you to create and manage WAF rules in a single location, simplifying the configuration process.
Automatic Deployment: Automatically deploys WAF rules to protected resources, reducing manual effort.
Policy-Based Control: Provides granular control over security policies, allowing you to tailor them to specific needs.
upvoted 2 times

leoo55 7 months, 4 weeks ago

upvoted 2 times

ChymKuBoy 7 months, 4 weeks ago

Selected Answer: B

B for sure
Centralized management: AWS Firewall Manager allows you to centrally manage AWS WAF rules across multiple accounts and regions. This simplifies the configuration and management process.

Consistent security policies: You can enforce consistent security policies across all your API Gateway APIs, ensuring that they are protected from the same threats.

Scalability: AWS Firewall Manager can handle a large number of accounts and resources, making it suitable for global companies with many API Gateway APIs.
upvoted 3 times

PaulGa 9 months ago

Selected Answer: A

Ans A - "With AWS WAF, you can create security rules that control bot traffic and block common attack patterns such as SQL injection or cross-site scripting (XSS). Use cases: Filter web traffic."

injection or cross-site scripting (XSS). Use cases: Filter web traffic.
https://aws.amazon.com › waf

None of the other options can do it.
upvoted 1 times

Americanman 9 months, 4 weeks ago

AWS WAF helps you to protect your application against common web exploits and bots that can affect availability, compromise security or consume excessive resources.
You can create security rules that will control bot traffic and common attacks like SQL injection or Cross-site scripting (XSS)
upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: B

A is valid, but B achieves the least operational overhead.
upvoted 1 times

TilTil 1 year, 2 months ago

Selected Answer: A

WAF deals well with the types of attacks mentioned. XSS and SQL Injection are both app level attacks hence needs a WAF.
upvoted 2 times

sirasdf 1 year, 3 months ago

B

Option A involves setting up AWS WAF in both regions and associating regional web ACLs with an API stage. While this can provide the necessary protection, it requires more manual configuration in each region, potentially leading to more administrative effort, especially if there are updates or changes needed to be made across multiple regions.

Therefore, Option B is likely to require the least amount of administrative effort.
upvoted 2 times

killbots 1 year, 4 months ago

Selected Answer: A

Original architecture does not have WAFs. B assumes there are WAFs already in place and why would you want to deploy a Firewall Manager to manage 1 Firewall? it adds unnecessary administrative tasks and costs for a tool that is not needed. You would want that if you were managing 10+ Firewalls not just one. A makes the most sense.
upvoted 4 times

thewalker 1 year, 4 months ago

Selected Answer: B

B is the answer
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #120 *Topic 1*

A company has implemented a self-managed DNS solution on three Amazon EC2 instances behind a Network Load Balancer (NLB) in the us-west-2 Region. Most of the company's users are located in the United States and Europe. The company wants to improve the performance and availability of the solution. The company launches and configures three EC2 instances in the eu-west-1 Region and adds the EC2 instances as targets for a new NLB.
Which solution can the company use to route traffic to all the EC2 instances?

A. Create an Amazon Route 53 geolocation routing policy to route requests to one of the two NLBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

B. Create a standard accelerator in AWS Global Accelerator. Create endpoint groups in us-west-2 and eu-west-1. Add the two NLBs as endpoints for the endpoint groups. **Most Voted**

C. Attach Elastic IP addresses to the six EC2 instances. Create an Amazon Route 53 geolocation routing policy to route requests to one of the six EC2 instances. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

D. Replace the two NLBs with two Application Load Balancers (ALBs). Create an Amazon Route 53 latency routing policy to route requests to one of the two ALBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

Correct Answer: *B*

Community vote distribution

B (79%) A (18%) D

Comments

dokaedu Highly Voted 2 years, 7 months ago

B is the correct one for seld manage DNS
If need to use Route53, ALB (layar 7) needs to be used as end points for 2 reginal x 3 EC2s, if it the case answer would be the option 4
upvoted 21 times

EllenLiu 6 months ago

A is not correct as should use NLB or ALB as the distribution origin?
upvoted 1 times

MutiverseAgent 1 year, 11 months ago

After reading the discussion I think the right answer is B, as the service they use is DNS it does not make sense using a cloudfront distribution for this. The scenario would be different if the service were HTTP/HTTPS.

upvoted 6 times

MutiverseAgent 1 year, 11 months ago

Just to complete my previous comment. If the scenario were that the company uses HTTP/HTTPS service, then the correct answer (as the original dokaedu message mentions) would be option D)

upvoted 4 times

RNess 1 year, 7 months ago

Why I need replace NLB to ALB?

upvoted 2 times

pentium75 1 year, 5 months ago

Who said that?

upvoted 3 times

LeGloupier Highly Voted 2 years, 7 months ago

Selected Answer: B

for me it is B

upvoted 13 times

satyaammm Most Recent 5 months, 1 week ago

Selected Answer: B

AWS Global Accelerator is the most suitable here as it provides low latency and makes the AWS Network closer.

upvoted 1 times

PaulGa 9 months ago

Selected Answer: B

Ans B - "AWS Global Accelerator improves the availability and performance of your applications for global users by routing traffic to the optimal endpoint based on performance and policies."

https://aws.amazon.com › global-accelerator › faqs

upvoted 2 times

rityoui 1 year, 3 months ago

Selected Answer: B

i choose a previous until i checked google that tells me "DNS is an Application-layer protocol"

upvoted 3 times

thewalker 1 year, 4 months ago

Selected Answer: A

A seems the right answer.

upvoted 1 times

pentium75 1 year, 5 months ago

Selected Answer: B

Not A: CloudFront is not for DNS
Not C: Involves CloudFront which is not needed, otherwise would work but ignore the NLBs
Not D: ALB can't handle DNS
Leaves B

upvoted 5 times

SaurabhTiwari1 1 year, 5 months ago

Selected Answer: B

Keyword-
AWS global accelerator = Super cop (who direct the traffic and give you the best way to reach your destination)

Geolocation is use for showing web content as you want to show your web content to particular country or continent.
Geolocation has nothing to do with traffic.

upvoted 3 times

SaurabhTiwari1 1 year, 5 months ago

Route 53 geolocation has nothing to do with traffic in the sense that it does not affect the amount or speed of traffic that reaches your resources. It only affects how Route 53 responds to DNS queries based on the location of your users.

upvoted 2 times

Masakichen 1 year, 6 months ago

Option B. Create a standard accelerator in AWS Global Accelerator. Establish endpoint groups in us-west-2 and eu-west-1. Add two NLBs as endpoints of the endpoint group.

AWS Global Accelerator is a network service that can provide a global traffic management solution. By creating a standard accelerator in AWS Global Accelerator, you can guide user traffic to the endpoint closest to them, thereby improving the performance and availability of the application. In this case, you can establish endpoint groups in the us-west-2 and eu-west-1 regions, and add two NLBs as endpoints. In this way, no matter where the user is located, their requests will be routed to the EC2 instance closest to them, thereby improving the performance and availability of DNS resolution. In addition, this design can also provide flexibility and scalability to handle a large amount of traffic. Therefore, this solution can meet your needs.

upvoted 8 times

Ruffyit 1 year, 7 months ago

Global Accelerator: AWS Global Accelerator is designed to improve the availability and performance of applications by using static IP addresses (Anycast IPs) and routing traffic over the AWS global network infrastructure.

Endpoint Groups: By creating endpoint groups in both the us-west-2 and eu-west-1 Regions, the company can effectively distribute traffic to the NLBs in both Regions. This improves availability and allows traffic to be directed to the closest Region based on latency.

upvoted 2 times

tom_cruise 1 year, 8 months ago

Selected Answer: B

Key: route traffic to all the EC2 instances

upvoted 2 times

Hassaoo 1 year, 9 months ago

B. Create a standard accelerator in AWS Global Accelerator. Create endpoint groups in us-west-2 and eu-west-1. Add the two NLBs as endpoints for the endpoint groups.

Here's why this option is the most suitable:

Global Accelerator: AWS Global Accelerator is designed to improve the availability and performance of applications by using static IP addresses (Anycast IPs) and routing traffic over the AWS global network infrastructure.

Endpoint Groups: By creating endpoint groups in both the us-west-2 and eu-west-1 Regions, the company can effectively distribute traffic to the NLBs in both Regions. This improves availability and allows traffic to be directed to the closest Region based on latency.

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

B is the best solution to route traffic to all the EC2 instances across regions.

The key reasons are:

AWS Global Accelerator allows routing traffic to endpoints in multiple AWS Regions. It uses the AWS global network to optimize availability and performance.
Creating an accelerator with endpoint groups in us-west-2 and eu-west-1 allows traffic to be distributed across both regions.
Adding the NLBs in each region as endpoints allows the traffic to be routed to the EC2 instances behind them.
This provides improved performance and availability compared to just using Route 53 geolocation routing.

upvoted 6 times

MNotABot 1 year, 11 months ago

B
route requests to one of the two NLBs --> hence AD out / Attach Elastic IP addresses --> who will pay for it?

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Option B offers a global solution by utilizing Global Accelerator. By creating a standard accelerator and configuring endpoint groups in both Regions, the company can route traffic to all the EC2 across multiple regions. Adding the two NLBs as endpoints ensures that traffic is distributed effectively.

Option A does not directly address the requirement of routing traffic to all EC2 instances. It focuses on routing based on geolocation and using CloudFront as a distribution, which may not achieve the desired outcome.

Option C involves managing Elastic IP addresses and routing based on geolocation. However, it may not provide the same level of performance and availability as AWS Global Accelerator.

Option D focuses on ALBs and latency-based routing. While it can be a valid solution, it does not utilize AWS Global Accelerator and may require more configuration and management compared to option B.

upvoted 4 times

beginnercloud 2 years ago

Selected Answer: B

Correctly is B.

if it is self-managed DNS, you cannot use Route 53. There can be only 1 DNS service for the domain.

upvoted 5 times

studynoplay 2 years, 1 month ago

Selected Answer: B

For self-managed DNS solution:
https://aws.amazon.com/blogs/security/how-to-protect-a-self-managed-dns-service-against-ddos-attacks-using-aws-global-accelerator-and-aws-shield-advanced/

upvoted 3 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #121 *Topic 1*

A company is running an online transaction processing (OLTP) workload on AWS. This workload uses an unencrypted Amazon RDS DB instance in a Multi-AZ deployment. Daily database snapshots are taken from this instance.
What should a solutions architect do to ensure the database and snapshots are always encrypted moving forward?

A. Encrypt a copy of the latest DB snapshot. Replace existing DB instance by restoring the encrypted snapshot. **Most Voted**

B. Create a new encrypted Amazon Elastic Block Store (Amazon EBS) volume and copy the snapshots to it. Enable encryption on the DB instance.

C. Copy the snapshots and enable encryption using AWS Key Management Service (AWS KMS) Restore encrypted snapshot to an existing DB instance.

D. Copy the snapshots to an Amazon S3 bucket that is encrypted using server-side encryption with AWS Key Management Service (AWS KMS) managed keys (SSE-KMS).

Correct Answer: *A*

Community vote distribution

A (85%) Other (15%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: A

"You can enable encryption for an Amazon RDS DB instance when you create it, but not after it's created. However, you can add encryption to an unencrypted DB instance by creating a snapshot of your DB instance, and then creating an encrypted copy of that snapshot. You can then restore a DB instance from the encrypted snapshot to get an encrypted copy of your original DB instance."
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/encrypt-an-existing-amazon-rds-for-postgresql-db-instance.html

upvoted 66 times

1e22522 10 months, 2 weeks ago

thats crazy cuh the more you know ig

upvoted 2 times

Futurebones 2 years ago

How can A gurantee future encryption?

How can A gurantee future encryption?
upvoted 4 times

Smart 1 year, 10 months ago

Once DB is encrypted, newer snapshots and read replicas will also be encrypted.
upvoted 8 times

JoeGuan 1 year, 10 months ago

I agree, there is no reason to copy all of the snapshots and ecnrypt them all. You just need one encrypted snapshot, moving forward they will all be encrypted. C is close but I think there is no reason to copy all the snapshots plural. There is a wizard to go through and select the snapshot to encrypt. "In the Amazon RDS console navigation pane, choose Snapshots, and select the DB snapshot you created. For Actions, choose Copy Snapshot. Provide the destination AWS Region and the name of the DB snapshot copy in the corresponding fields. Select the Enable Encryption checkbox. For Master Key, specify the KMS key identifier to use to encrypt the DB snapshot copy. Choose Copy Snapshot. For more information, see Copying a snapshot in the Amazon RDS documentation". What if you had 30 snapshotS? You just need to do it once.
upvoted 4 times

Guru4Cloud 1 year, 10 months ago

In simple terms, you double it the affort of your work and spending money by creating unnessary snapshots... so A is the best choice
upvoted 2 times

kruasan Highly Voted 2 years, 1 month ago

Selected Answer: A

You can't restore from a DB snapshot to an existing DB instance; a new DB instance is created when you restore.
upvoted 5 times

rmanuraj Most Recent 5 months, 4 weeks ago

Selected Answer: A

A seems to be the correct option compared to C. In the question it clearly says the current DB instance is an unencrypted Amazon RDS DB instance. And you can't restore an encrypted snapshot to an unencrypted DB instance.
upvoted 1 times

PaulGa 8 months, 4 weeks ago

Selected Answer: A

Ans A - I must admit almsot put Ans C, but re-reading question and seeing comments its clear that encryption is needed "moving forward" so C is overkill...
upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

Replacing a snapshots creates a new one instead of restoring the old one.
upvoted 2 times

Saadiii 1 year, 1 month ago

Selected Answer: A

I feel this is a bit tricky in the way the question is asked, but C implies that you are encrypting the snapshot. You are not. It is the DB that receives a KMS key upon restoring, but the snapshot is still unencrypted
upvoted 2 times

SinghJagdeep 1 year, 5 months ago

Selected Answer: A

Correct. Please visit for more details.
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/encrypt-an-existing-amazon-rds-for-postgresql-db-instance.html
upvoted 2 times

ansagr 1 year, 6 months ago

Selected Answer: A

AWS RDS does not support direct restoration of an encrypted snapshot to an existing unencrypted DB instance. When you restore a snapshot, it creates a new DB instance with the same configuration as the original instance.
upvoted 2 times

tom_cruise 1 year, 7 months ago

Selected Answer: A

What's wrong with C is: "Copy the snapshots and enable encryption"
upvoted 3 times

tom_cruise 1 year, 8 months ago

Selected Answer: A

key: snapshots
upvoted 2 times

AntonioMinolfi 1 year, 8 months ago

Selected Answer: A

I was undecided if to choose A or C.
But since you can't restore a snapshot to an existing instance C is out. You can only create a new one.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_RestoreFromSnapshot.html#:~:text=You%20can%27t%20restore%20from%20a%20DB%20snapshot%20to%20an%20existing%20DB%20instance%3B%20a%20new%20DB%20instance%20is%20created%20when%20you%20restore.
upvoted 2 times

TMabs 1 year, 8 months ago

A makes sence
upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: C

A. Replacing the existing DB instance with an encrypted snapshot can result in downtime and potential data loss during migration.

B. Creating a new encrypted EBS volume for snapshots does not address the encryption of the DB instance itself.

D. Copying snapshots to an encrypted S3 bucket only encrypts the snapshots, but does not address the encryption of the DB instance.

Option C is the most suitable as it involves copying and encrypting the snapshots using AWS KMS, ensuring encryption for both the database and snapshots.
upvoted 2 times

BartoszGolebiowski24 1 year, 7 months ago

From the question:
"...What should a solutions architect do to ensure the database and snapshots are always encrypted moving forward?"
I think the question is about encrypting current and future snapshots instead of the old snapshots.
upvoted 2 times

Abrar2022 2 years ago

If daily snapshots are taken from the daily DB instance. Why create another copy? You just need to encrypt the latest daily DB snapshot and the restore from the existing encrypted snapshot.
upvoted 3 times

C_M_M 2 years, 1 month ago

A and C are almost similar except that A is latest snapshot, while C is snapshots (all the snapshots).
I don't see any other difference btw those two options.
Option A is clearly the correct on as all you need is the latest snapshot.
upvoted 3 times

JoeGuan 1 year, 10 months ago

I agree, in the wizard you would select ONE SNAPSHOT (singular in A), not all of the SNAPSHOTS (Plural in C)

upvoted 1 times

rushlav 2 years, 2 months ago

A
You can only encrypt an Amazon RDS DB instance when you create it, not after the DB instance is created.
However, because you can encrypt a copy of an unencrypted snapshot, you can effectively add encryption to an unencrypted DB instance. That is, you can create a snapshot of your DB instance, and then create an encrypted copy of that snapshot. You can then restore a DB instance from the encrypted snapshot, and thus you have an encrypted copy of your original DB instance.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.Encryption.html

upvoted 2 times

Abhineet9148232 2 years, 2 months ago

Selected Answer: C

Encryption is enabled during the Copy process itself.
https://repost.aws/knowledge-center/encrypt-rds-snapshots

upvoted 1 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #122 *Topic 1*

A company wants to build a scalable key management infrastructure to support developers who need to encrypt data in their applications.
What should a solutions architect do to reduce the operational burden?

A. Use multi-factor authentication (MFA) to protect the encryption keys.

B. Use AWS Key Management Service (AWS KMS) to protect the encryption keys. **Most Voted**

C. Use AWS Certificate Manager (ACM) to create, store, and assign the encryption keys.

D. Use an IAM policy to limit the scope of users who have access permissions to protect the encryption keys.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: B

If you are a developer who needs to digitally sign or verify data using asymmetric keys, you should use the service to create and manage the private keys you'll need. If you're looking for a scalable key management infrastructure to support your developers and their growing number of applications, you should use it to reduce your licensing costs and operational burden...
https://aws.amazon.com/kms/faqs/#:~:text=If%20you%20are%20a%20developer%20who%20needs%20to%20digitally,a%20broad%20set%20of%20industry%20and%20regional%20compliance%20regimes.

upvoted 24 times

ocbn3wby 2 years, 6 months ago

Most documented answers. Thank you, 123jhl0.

upvoted 5 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: B

The correct answer is Option B. To reduce the operational burden, the solutions architect should use AWS Key Management Service (AWS KMS) to protect the encryption keys.

AWS KMS is a fully managed service that makes it easy to create and manage encryption keys. It allows developers to easily encrypt and decrypt data in their applications, and it automatically handles the underlying key management tasks, such as key generation, key rotation, and key deletion. This can help to reduce the operational burden associated with key management.

upvoted 8 times

PaulGa Most Recent 8 months, 4 weeks ago

Selected Answer: B

Ans B - AWS KMS does it all as a managed service...

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: B

to reduce the operational burden, option B is the best choice.

upvoted 2 times

Dani29 1 year, 1 month ago

Selected Answer: B

The correct answer is Option B. To reduce the operational burden, the solutions architect should use AWS Key Management Service (AWS KMS) to protect the encryption keys

upvoted 2 times

Ruffyit 1 year, 7 months ago

AWS KMS handles the encryption key management, rotation, and auditing. This removes the undifferentiated heavy lifting for developers.
KMS integrates natively with many AWS services like S3, EBS, RDS for encryption. This makes it easy to encrypt data.
KMS scales automatically as key usage increases. Developers don't have to worry about provisioning key infrastructure.
Fine-grained access controls are available via IAM policies and grants. KMS is secure by default.
Features like envelope encryption make compliance easier for regulated workloads.
AWS handles the hardware security modules (HSMs) for cryptographic key storage

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

The main reasons are:

AWS KMS handles the encryption key management, rotation, and auditing. This removes the undifferentiated heavy lifting for developers.
KMS integrates natively with many AWS services like S3, EBS, RDS for encryption. This makes it easy to encrypt data.
KMS scales automatically as key usage increases. Developers don't have to worry about provisioning key infrastructure.
Fine-grained access controls are available via IAM policies and grants. KMS is secure by default.
Features like envelope encryption make compliance easier for regulated workloads.
AWS handles the hardware security modules (HSMs) for cryptographic key storage

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: B

By utilizing AWS KMS, the company can offload the operational responsibilities of key management, including key generation, rotation, and protection. AWS KMS provides a scalable and secure infrastructure for managing encryption keys, allowing developers to easily integrate encryption into their applications without the need to manage the underlying key infrastructure.

Option A (MFA), option C (ACM), and option D (IAM policy) are not directly related to reducing the operational burden of key management. While these options may provide additional security measures or access controls, they do not specifically address the scalability and management aspects of a key management infrastructure. AWS KMS is designed to simplify the key management process and is the most suitable option for reducing the operational burden in this scenario.

upvoted 3 times

cheese929 2 years, 1 month ago

Selected Answer: B

B is correct.

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B
upvoted 1 times

Wpcorgan 2 years, 6 months ago

B is correct
upvoted 1 times

Wpcorgan 2 years, 6 months ago

B is correct
upvoted 1 times

Jtic 2 years, 7 months ago

Selected Answer: B

If you are responsible for securing your data across AWS services, you should use it to centrally manage the encryption keys that control access to your data. If you are a developer who needs to encrypt data in your applications, you should use the AWS Encryption SDK with AWS KMS to easily generate, use and protect symmetric encryption keys in your code.
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #123 *Topic 1*

A company has a dynamic web application hosted on two Amazon EC2 instances. The company has its own SSL certificate, which is on each instance to perform SSL termination.
There has been an increase in traffic recently, and the operations team determined that SSL encryption and decryption is causing the compute capacity of the web servers to reach their maximum limit.
What should a solutions architect do to increase the application's performance?

A. Create a new SSL certificate using AWS Certificate Manager (ACM). Install the ACM certificate on each instance.

B. Create an Amazon S3 bucket Migrate the SSL certificate to the S3 bucket. Configure the EC2 instances to reference the bucket for SSL termination.

C. Create another EC2 instance as a proxy server. Migrate the SSL certificate to the new instance and configure it to direct connections to the existing EC2 instances.

D. Import the SSL certificate into AWS Certificate Manager (ACM). Create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM. Most Voted

Correct Answer: *D*

Community vote distribution

D (96%) A

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D

This issue is solved by SSL offloading, i.e. by moving the SSL termination task to the ALB.
https://aws.amazon.com/blogs/aws/elastic-load-balancer-support-for-ssl-termination/
upvoted 23 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: D

The correct answer is D. To increase the application's performance, the solutions architect should import the SSL certificate into AWS Certificate Manager (ACM) and create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM.

An Application Load Balancer (ALB) can offload the SSL termination process from the EC2 instances, which can help to increase the compute capacity available for the web application. By creating an ALB with an HTTPS listener and using the SSL certificate the compute capacity available for the web application. By creating an ALB with an HTTPS listener and using the SSL certificate from ACM, the ALB can handle the SSL termination process, leaving the EC2 instances free to focus on running the web application.

upvoted 14 times

satyaammm Most Recent 5 months ago

Selected Answer: D

Since the company has its own SSL certificate so its more suitable to import the company's SSL certificate and using HTTPS.

upvoted 1 times

PaulGa 8 months, 4 weeks ago

Ans D - well explained by Buruguduystunstugudunstuy (1yr, 8mth):
"To increase the application's performance, the solutions architect should import the SSL certificate into AWS Certificate Manager (ACM) and create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM.

An Application Load Balancer (ALB) can offload the SSL termination process from the EC2 instances, which can help to increase the compute capacity available for the web application. By creating an ALB with an HTTPS listener and using the SSL certificate from ACM, the ALB can handle the SSL termination process, leaving the EC2 instances free to focus on running the web application."

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

Ruffyit 1 year, 7 months ago

This issue is solved by SSL offloading, i.e. by moving the SSL termination task to the ALB.
https://aws.amazon.com/blogs/aws/elastic-load-balancer-support-for-ssl-termination

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The key reasons are:

Using an Application Load Balancer with an HTTPS listener allows SSL termination to happen at the load balancer layer.
The EC2 instances behind the load balancer receive only unencrypted traffic, reducing load on them.
Importing the custom SSL certificate into ACM allows the ALB to use it for HTTPS listeners.
This removes the need to install and manage SSL certificates on each EC2 instance.
ALB handles the SSL overhead and scales automatically. The EC2 fleet focuses on app logic.
Options A, B, C don't offload SSL overhead from the EC2 instances themselves.

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: D

By using ACM to manage the SSL certificate and configuring an ALB with HTTPS listener, the SSL termination will be handled by the load balancer instead of the web servers. This offloading of SSL processing to the ALB reduces the compute capacity burden on the web servers and improves their performance by allowing them to focus on serving the dynamic web application.

Option A suggests creating a new SSL certificate using ACM, but it does not address the SSL termination offloading and load balancing capabilities provided by an ALB.

Option B suggests migrating the SSL certificate to an S3 bucket, but this approach does not provide the necessary SSL termination and load balancing functionalities.

Option C suggests creating another EC2 instance as a proxy server, but this adds unnecessary complexity and management overhead without leveraging the benefits of ALB's built-in load balancing and SSL termination capabilities.

Therefore, option D is the most suitable choice to increase the application's performance in this scenario.

upvoted 3 times

dejung 2 years, 4 months ago

Selected Answer: A

Why is A wrong?
upvoted 2 times

Yadav_Sanjay 2 years, 1 month ago

Company uses its own SSL certificate. Option A says.. Create a SSL certificate in ACM
upvoted 4 times

xdkonorek2 1 year, 7 months ago

ec2 instances still would be responsible for decrypting traffic and it wouldn't solve load issue
upvoted 2 times

remand 2 years, 4 months ago

Selected Answer: D

SSL termination is the process of ending an SSL/TLS connection. This is typically done by a device, such as a load balancer or a reverse proxy, that is positioned in front of one or more web servers. The device decrypts incoming SSL/TLS traffic and then forwards the unencrypted request to the web server. This allows the web server to process the request without the overhead of decrypting and encrypting the traffic. The device then re-encrypts the response from the web server and sends it back to the client. This allows the device to offload the SSL/TLS processing from the web servers and also allows for features such as SSL offloading, SSL bridging, and SSL acceleration.
upvoted 5 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D to offload the SSL encryption workload
upvoted 2 times

Aamee 2 years, 6 months ago

Selected Answer: D

Due to this statement particularly: "The company has its own SSL certificate" as it's not created from AWS ACM itself.
upvoted 1 times

Wpcorgan 2 years, 6 months ago

D is correct
upvoted 1 times

Six_Fingered_Jose 2 years, 7 months ago

Selected Answer: D

agree with D
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #124 *Topic 1*

A company has a highly dynamic batch processing job that uses many Amazon EC2 instances to complete it. The job is stateless in nature, can be started and stopped at any given time with no negative impact, and typically takes upwards of 60 minutes total to complete. The company has asked a solutions architect to design a scalable and cost-effective solution that meets the requirements of the job.
What should the solutions architect recommend?

A. Implement EC2 Spot Instances. **Most Voted**

B. Purchase EC2 Reserved Instances.

C. Implement EC2 On-Demand Instances.

D. Implement the processing on AWS Lambda.

Correct Answer: *A*

Community vote distribution

A (96%) Other

Comments

Kapello10 Highly Voted 2 years, 6 months ago

Selected Answer: A

Cant be implemented on Lambda because the timeout for Lambda is 15mins and the Job takes 60minutes to complete

Answer >> A
upvoted 25 times

Evangelia Highly Voted 2 years, 7 months ago

spot instances
upvoted 6 times

Kazmin Most Recent 2 months, 1 week ago

Selected Answer: B

Spot is a bad choice because process takes up to 60 mins and instances can be down at any time
upvoted 1 times

Neil_12345678 3 months, 3 weeks ago

Selected Answer: D

Ec2 by themselves are not "scalable" unless they are under an autoscaling group.The question also says that the job can be started and stopped with no negative impact.So even if the execution time of 15mins runs out, it can be started again to complete the 60 mins job. Hence my answer is D

upvoted 1 times

jaradat02 10 months, 3 weeks ago

Selected Answer: A

A is the correct answer, we can't use lambda because it's limit is 15 mins.

upvoted 3 times

thewalker 1 year, 4 months ago

Selected Answer: A

There is a chance of interrupting the jobs, but as they can be started and stopped at any given time, the MOST COST effective is going for Spot.

upvoted 4 times

Ruffyit 1 year, 7 months ago

Spot Instances provide significant cost savings for flexible start and stop batch jobs.
Purchasing Reserved Instances (B) is better for stable workloads, not dynamic ones.
On-Demand Instances (C) are costly and lack potential cost savings like Spot Instances.
AWS Lambda (D) is not suitable for long-running batch jobs.

upvoted 3 times

tom_cruise 1 year, 8 months ago

Selected Answer: A

key: can be started and stopped at any given time with no negative impact

upvoted 4 times

AbhilashDyadav 1 year, 8 months ago

Selected Answer: A

Spot can do that

upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

The key reasons are:

Spot can provide significant cost savings (up to 90%) compared to On-Demand.
Since the job is stateless and can be stopped/restarted anytime, the intermittent availability of Spot is not an issue.
Spot supports the same instance types as On-Demand, so optimal instance types can be chosen.
For a 60+ minute batch job, the chance of Spot interruption is low. But if it happens, the job can just be restarted.
Reserved Instances don't offer any advantage for a highly dynamic job like this.
Lambda is not a good fit given the long runtime requirement.

upvoted 4 times

cookieMr 1 year, 11 months ago

Selected Answer: A

Spot Instances provide significant cost savings for flexible start and stop batch jobs.
Purchasing Reserved Instances (B) is better for stable workloads, not dynamic ones.
On-Demand Instances (C) are costly and lack potential cost savings like Spot Instances.
AWS Lambda (D) is not suitable for long-running batch jobs.

upvoted 1 times

beginnercloud 2 years ago

Selected Answer: A

A is correct

upvoted 1 times

alexiscloud 2 years, 2 months ago

Answer A:
typically takes upwards of 60 minutes total to complete.

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: A

The correct answer is Option A. To design a scalable and cost-effective solution for the batch processing job, the solutions architect should recommend implementing EC2 Spot Instances.

EC2 Spot Instances allow users to bid on spare Amazon EC2 computing capacity and can be a cost-effective solution for stateless, interruptible workloads that can be started and stopped at any time. Since the batch processing job is stateless, can be started and stopped at any time, and typically takes upwards of 60 minutes to complete, EC2 Spot Instances would be a good fit for this workload.

upvoted 2 times

k1kavi1 2 years, 5 months ago

Selected Answer: A

Spot Instances should be good enough and cost effective because the job can be started and stopped at any given time with no negative impact.

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: A

Option A

upvoted 1 times

Wpcorgan 2 years, 6 months ago

A is correct

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #125 *Topic 1*

A company runs its two-tier ecommerce website on AWS. The web tier consists of a load balancer that sends traffic to Amazon EC2 instances. The database tier uses an Amazon RDS DB instance. The EC2 instances and the RDS DB instance should not be exposed to the public internet. The EC2 instances require internet access to complete payment processing of orders through a third-party web service. The application must be highly available.
Which combination of configuration options will meet these requirements? (Choose two.)

A. Use an Auto Scaling group to launch the EC2 instances in private subnets. Deploy an RDS Multi-AZ DB instance in private subnets. **Most Voted**

B. Configure a VPC with two private subnets and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the private subnets.

C. Use an Auto Scaling group to launch the EC2 instances in public subnets across two Availability Zones. Deploy an RDS Multi-AZ DB instance in private subnets.

D. Configure a VPC with one public subnet, one private subnet, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnet.
D. Configure a VPC with two public subnets, two private subnets, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnets. **Most Voted**

Correct Answer: *AD*

Community vote distribution

AD (61%) | AB (18%) | A (17%) | Other

Comments

HayLLlHuK Highly Voted 2 years, 5 months ago

A and E!
Application has to be highly available while the instance and database should not be exposed to the public internet, but the instances still requires access to the internet. NAT gateway has to be deployed in public subnets in this case while instances and database remain in private subnets in the VPC, therefore answer is (A) and (E).
https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html

If the instances did not require access to the internet, then the answer could have been
(B) to use a private NAT gateway and keep it in the private subnets to communicate only to the VPCs.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Scenario2.html

upvoted 30 times

darn 2 years, 1 month ago

your link is right but your voting is wrong, should be AD, although that still doesnt explain why 2 NAT gateways

upvoted 3 times

JA2018 6 months, 4 weeks ago

It looks like there could be a typo error in the list of option. HayLLiHuK is referring to Option 'E' which is missing.

upvoted 1 times

cheroh_tots 1 year, 3 months ago

Because NAT gateways are availability zone specific, if you need HA you will need a NAT gateway in each availability zone.

upvoted 1 times

ale_brd_111 1 year, 6 months ago

cus application has to be HA, if one NAT gateway fails the other could take the traffic

upvoted 4 times

mabotega Highly Voted 2 years, 7 months ago

Selected Answer: AD

Answer A for: The EC2 instances and the RDS DB instance should not be exposed to the public internet. Answer D for: The EC2 instances require internet access to complete payment processing of orders through a third-party web service. Answer A for: The application must be highly available.

upvoted 28 times

oguzbeliren 1 year, 10 months ago

D allows public internet access which is not desired. The answer is not d.
The most accurate answers are AB

upvoted 2 times

pentium75 1 year, 5 months ago

B is wrong because you can't deploy NAT GW in a private subnet. Correct answer is E (mislabelled as a second D). Stem says that the EC2 instances (!) must not be exposed to the Internet, the Load Balancer can be exposed.

upvoted 3 times

AbhiJo 2 years, 6 months ago

We will require 2 private subnets, D does mention 1 subnet

upvoted 4 times

OluwaZettai 5 months, 2 weeks ago

Why do we require 2?

upvoted 1 times

pentium75 1 year, 5 months ago

There's two D options ;) second is correct

upvoted 4 times

smd_ 2 years, 1 month ago

why not option B.The EC2 instances can be launched in private subnets across two Availability Zones, and the Application Load Balancer can be deployed in the private subnets. NAT gateways can be configured in each private subnet to provide internet access for the EC2 instances to communicate with the third-party web service.

upvoted 2 times

ruqui 2 years ago

B option wrong! NAT gateways must be created in public subnets!!

upvoted 8 times

x33 1 year, 9 months ago

I think you are wrong on this. In fact, NAT gateways are typically created in private subnets.
upvoted 3 times

RNess 1 year, 7 months ago

NAT Gateway can’t be used by EC2 instance in the same subnet (only from other subnets)
upvoted 4 times

bora4motion Most Recent 1 month, 2 weeks ago

Selected Answer: A

Correct the answers already! Last option is E not D!
AE
upvoted 1 times

CloudExpert01 2 months, 1 week ago

Selected Answer: AD

If D is chosen: An additional private subnet in the second Availability Zone is needed for high availability.
upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: AD

Theright one is A and D. The last option (with numbering incorrectly shown as the second "D") is not correct, since a load balancer is not per AZ, rather for the entire VPC.
upvoted 1 times

mzeynalli 7 months, 3 weeks ago

Selected Answer: AD

AE
The goal is to create a highly available architecture with EC2 instances and RDS instances that are not exposed to the public internet, while still allowing the EC2 instances to access external services.
upvoted 1 times

PaulGa 8 months, 4 weeks ago

Selected Answer: AD

And A,D - keeps EC2 and RDS private and highly available, but with public front-end

(Not sure why Author's highlighted answer is C?)
upvoted 1 times

Shailesh1717 9 months ago

AD are correct ans
upvoted 1 times

scaredSquirrel 10 months, 1 week ago

AE
A because the EC2 can't be exposed to public
E because each subnet must reside entirely within one Availability Zone and cannot span zones.

source: https://docs.aws.amazon.com/vpc/latest/userguide/configure-subnets.html#:~:text=Delete%20a%20subnet-,Subnet%20basics,of%20a%20single%20Availability%20Zone.
upvoted 1 times

jaradat02 10 months, 3 weeks ago

A and D is the best combination, it achieves all of the desired requirements.
upvoted 1 times

shil_31 1 year ago

Selected Answer: AD

Option A ensures that EC2 instances and RDS DB instance are not exposed to the public internet, as they are launched in private subnets. Auto Scaling group will also ensure high availability of EC2 instances.
Option D configures a VPC with a public subnet for the load balancer, and a private subnet for the EC2 instances and RDS DB instance. NAT gateways in both Availability Zones will allow EC2 instances to access the internet for payment processing, while keeping them private.

upvoted 3 times

lofzee 1 year ago

A and E (the second D option) .

upvoted 1 times

Yash2804 1 year ago

There might be error in question. i modified it now CE answer seems to be correct

A company runs its two-tier ecommerce website on AWS. The web tier consists of a load balancer that sends traffic to Amazon EC2 instances. The database tier uses an Amazon RDS DB instance. The RDS DB instance should not be exposed to the public internet. The The RDS DB instance require internet access to complete payment processing of orders through a third-party web service. The application must be highly available.

Which combination of configuration options will meet these requirements? (Choose two.)

upvoted 1 times

Solomon2001 1 year, 1 month ago

Selected Answer: AB

Use an Auto Scaling group to launch the EC2 instances in private subnets. Deploy an RDS Multi-AZ DB instance in private subnets.
This ensures that the EC2 instances and the RDS DB instance are not exposed to the public internet.
Configure a VPC with two private subnets and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the private subnets.
This allows the EC2 instances in private subnets to access the internet for payment processing through the NAT gateways while keeping them private.

upvoted 2 times

EMPERBACH 1 year, 1 month ago

App layer & DB Layer does not expose to Internet -> both in private subnet, access through NAT Gateway
It's enough for all requirements!

upvoted 1 times

SaurabhTiwari1 1 year, 5 months ago

Selected Answer: AD

AD is right , last one D

upvoted 1 times

rlamberti 1 year, 7 months ago

Selected Answer: AD

AE
Two public subnets = two addresses for ALB = high availability
two private subnets with NAT gateway to allow eggress traffic to internet - application tier will be able to complete payment

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #126 *Topic 1*

A solutions architect needs to implement a solution to reduce a company's storage costs. All the company's data is in the Amazon S3 Standard storage class. The company must keep all data for at least 25 years. Data from the most recent 2 years must be highly available and immediately retrievable.
Which solution will meet these requirements?

A. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive immediately.

B. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 2 years. **Most Voted**

C. Use S3 Intelligent-Tiering. Activate the archiving option to ensure that data is archived in S3 Glacier Deep Archive.

D. Set up an S3 Lifecycle policy to transition objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately and to S3 Glacier Deep Archive after 2 years.

Correct Answer: *B*

Community vote distribution

B (83%) Other (17%)

Comments

rjam Highly Voted 2 years, 6 months ago

Selected Answer: B

Why Not C? Because Intelligent Tier the objects are automatically moved to different tiers.
The question says "the data from most recent 2 yrs should be highly available and immediately retrievable", which means in intelligent tier , if you activate archiving option(as Option C specifies) , the objects will be moved to Archive tiers(instant to access to deep archive access tiers) in 90 to 730 days. Remember these archive tiers performance will be similar to S3 glacier flexible and s3 deep archive which means files cannot be retrieved immediately within 2 yrs .

We have a hard requirement in question which says it should be retreivable immediately for the 2 yrs. which cannot be acheived in Intelligent tier. So B is the correct option imho.

Because of the above reason Its possible only in S3 standard and then configure lifecycle configuration to move to S3 Glacier Deep Archive after 2 yrs.

upvoted 17 times

MutiverseAgent 1 year, 11 months ago

Mmm.. You can enable Intelligent-Tiering and take advantage of of the infrequent Access tier and thus reducing costs. To avoid moving objects to the deep archive tier before the two years it would be enough to enable ONLY the check "Deep

Archive Access tier" and set days to 720 (two years, which is curiously the maximum value), and keep disabled the check "Archive Access tier" to avoid the Intelligent-Tiering move objects to the non-instant retrieval tier. That will work, offcourse this specific configuration is not mention in the question which leaves some doubts about which option is the correct.
upvoted 2 times

MutiverseAgent 1 year, 11 months ago

Just to clarify, my previous comment is about how answer B) might be correct and the MOST cheapest option under the correct configuration.
upvoted 2 times

MutiverseAgent 1 year, 11 months ago

Sorry, I meant answer C) might be correct
upvoted 1 times

Abdou1604 1 year, 9 months ago

but your S3 intelligent-tiering will move the object to S3 infrequent access tier which a is a single AZ tier , and then the HA requirement will not be respected
upvoted 2 times

sandordini 1 year, 1 month ago

S3 Standard IA is NOT single AZ. (One-Zone IA is single Az.)
upvoted 2 times

sandordini 1 year, 1 month ago

The issue is with the missing "flag" for 2 years, and not S3 Intelligent Tiering. It needs to be B.
upvoted 2 times

TelaO Highly Voted 2 years, 6 months ago

Selected Answer: B

B is the only right answer. C does not indicate archiving after 2 years. If it did specify 2 years, then C would also be an option.
upvoted 9 times

satyaammm Most Recent 5 months ago

Selected Answer: B

S3 Lifecycle policies works best here as we no longer need frequently access to data that has been over 2 years and hence moving it to S3 Glacier Deep Archive help reduce costs.
upvoted 1 times

PaulGa 8 months, 4 weeks ago

Selected Answer: B

Ans B - I did initially think Ans C, but rjam (1 yr, 10 mth ago) quickly quashed that notion:
"Why Not C? Because Intelligent Tier the objects are automatically moved to different tiers. The question says "the data from most recent 2 yrs should be highly available and immediately retrievable", which means in intelligent tier , if you activate archiving option(as Option C specifies) , the objects will be moved to Archive tiers(instant to access to deep archive access tiers) in 90 to 730 days. Remember these archive tiers performance will be similar to S3 glacier flexible and s3 deep archive which means files cannot be retrieved immediately within 2 yrs."
upvoted 1 times

bignatov 9 months, 2 weeks ago

Selected Answer: B

B is correct, because it fits to the requirements and it still cheaper than the option C.
upvoted 1 times

JaegEr_2k1 10 months, 2 weeks ago

Stupid question:
A: No immediately retrievable and cost
B: No immediately retrievable
C: Not unpredicted access

D: Immediately retrievable but not HA
F*ck the guy who made this question
upvoted 3 times

jaradat02 10 months, 3 weeks ago

Selected Answer: B

B is the correct choice, the requirements are very clear, intelligent tiering is used only when we don't have a clear pattern for the access of the data, when it's unpredictable.
upvoted 2 times

LoXoL 1 year, 5 months ago

Selected Answer: B

A. We can't move to Glacier immediately as data from last 2 yrs need to be immediately retrievable
B. It's the perfect fit: getting HA and instant access (with current solution = S3 std), then moving to Deep Archive after 2 yrs (very cheap)
C: Highly expensive because of Intelligent Tiering
D: it lacks HA with One Zone
upvoted 2 times

SaurabhTiwari1 1 year, 5 months ago

Selected Answer: B

Data remain in S3 standard storage for 2 years then it will be move to s3 glacier deep archive after 2 year.
upvoted 3 times

Ruffyit 1 year, 7 months ago

but your S3 intelligent-tiering will move the object to S3 infrequent access tier which a is a single AZ tier , and then the HA requirement will not be respected
upvoted 1 times

David_Ang 1 year, 8 months ago

Selected Answer: B

i understand why "B" is more correct than "C" and is because "C" is bad formulated, if in the answer would say "life cycle after 2 years of using intelligent tiring" then it would be the correct answer. so "B" is correct
upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

I would not opt for C simply because S3IT was specifically designed for scenarios where the access patterns are unknown.
This scenario has clearly known access patterns making option B the best.
upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Option A is incorrect because immediately transitioning objects to S3 Glacier Deep Archive would not fulfill the requirement of keeping the most recent 2 years of data highly available and immediately retrievable.

Option C is also incorrect because using S3 Intelligent-Tiering with archiving option would not meet the requirement of immediately retrievable data for the most recent 2 years.

Option D is not the best choice because transitioning objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) and then to S3 Glacier Deep Archive would not satisfy the requirement of immediately retrievable data for the most recent 2 years.

Option B is the correct solution. By setting up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 2 years, the company can keep all data for at least 25 years while ensuring that data from the most recent 2 years remains highly available and immediately retrievable in the Amazon S3 Standard storage class. This solution optimizes storage costs by leveraging the Glacier Deep Archive for long-term storage.
upvoted 3 times

kambarami 1 year, 9 months ago

this makes sense the question is a bit tricky. I now uderstand that all the data is already kept in S3 Standard meaning immediate retrieval of the most recent data is remains highly available

immediate retrieval of the most recent data is remains highly available.
upvoted 2 times

Yadav_Sanjay 1 year, 12 months ago

Why not D
upvoted 2 times

RNess 1 year, 7 months ago

"Data from the most recent 2 years must be highly available and immediately retrievable."
upvoted 1 times

RNess 1 year, 7 months ago

Additionally,
S3 Standard Availability = 99.99%
S3 One Zone-IA Availability = 99.5%
upvoted 2 times

Robrobtutu 2 years, 1 month ago

Selected Answer: B

B is the only one possible.
upvoted 1 times

rushlav 2 years, 2 months ago

C would not work as the names of these S3 archives are called Archive Access Tier and Deep Archive access tiers, so since they mention glacier in option C , I think its B which is the correct.
upvoted 2 times

CaoMengde09 2 years, 4 months ago

It's pretty straight forward.

S3 Standard answers for High Availaibility/Immediate retrieval for 2 years. S3 Intelligent Tiering would just incur additional cost of analysis while the company insures that it requires immediate retrieval in any moment and without risk to Availability. So a capital B
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #127 *Topic 1*

A media company is evaluating the possibility of moving its systems to the AWS Cloud. The company needs at least 10 TB of storage with the maximum possible I/O performance for video processing, 300 TB of very durable storage for storing media content, and 900 TB of storage to meet requirements for archival media that is not in use anymore.
Which set of services should a solutions architect recommend to meet these requirements?

A. Amazon EBS for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage

B. Amazon EBS for maximum performance, Amazon EFS for durable data storage, and Amazon S3 Glacier for archival storage

C. Amazon EC2 instance store for maximum performance, Amazon EFS for durable data storage, and Amazon S3 for archival storage

D. Amazon EC2 instance store for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage **Most Voted**

Correct Answer: *D*

Community vote distribution

D (57%) A (42%) B

Comments

Sauran Highly Voted 2 years, 7 months ago

Selected Answer: D

Max instance store possible at this time is 30TB for NVMe which has the higher I/O compared to EBS.

is4gen.8xlarge 4 x 7,500 GB (30 TB) NVMe SSD

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html#instance-store-volumes

upvoted 39 times

Tsige 7 months, 4 weeks ago

Answer is A
While instance store provides fast local storage, it is ephemeral (data is lost when the instance stops or terminates). It's not suitable for workloads that require persistence, like video processing that may span across multiple EC2 instances over time. EBS provides persistent, high-performance storage.

upvoted 7 times

JA2018 6 months, 4 weeks ago

err, stem did not mention requirement for durable storage for the 10TB used for video processing....
upvoted 2 times

ishitamodi4 2 years, 5 months ago

instance store volume for the root volume, the size of this volume varies by AMI, but the maximum size is 10 GB
upvoted 1 times

JayBee65 2 years, 5 months ago

This link shows a max capacity of 30TB, so what is the problem?
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html#instance-store-volumes
upvoted 2 times

JayBee65 2 years, 5 months ago

Only the following instance types support an instance store volume as the root device: C3, D2, G2, I2, M3, and R3, and we're using an I3, so an instance store volume is irrelevant.
upvoted 5 times

antropaws 2 years ago

THE CORRECT ANSWER IS A.

The biggest Instance Store Storage Optimized option (is4gen.8xlarge) has a capacity of only 3TB.

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instance-store-volumes.html#instance-store-vol-so
upvoted 3 times

michellemeloc 2 years, 1 month ago

Update: i3en.metal and i3en.24xlarge = 8 x 7500 GB (60TB)
upvoted 3 times

MiniYang Highly Voted 1 year, 6 months ago

Selected Answer: A

The correct Answer is A :
Amazon EC2 instance store (Instance Store) is usually not the best choice because the storage it provides is temporary and tied to the life cycle of the instance. When an instance is stopped or terminated, data on the instance store is lost.

In this scenario, the company's requirements were to have the maximum possible I/O performance and required durable data storage. Therefore, using Amazon EC2 Instance Store does not meet these requirements because it lacks durability.

In contrast, Amazon EBS (Elastic Block Store) provides persistent regional block storage and can meet the needs of high-performance I/O. Therefore, the answer should include Amazon EBS, not Amazon EC2 instance storage.
upvoted 26 times

pentium75 1 year, 5 months ago

"The company's requirements were to have the maximum possible I/O performance and required durable data storage." Yeah, but not for the same data.

10 TB "maximum possible I/O performance" for processing (= temporary)
300 TB "very durable" (= S3)
900 TB "for archival" (= Glacier)
upvoted 14 times

LoXoL 1 year, 5 months ago

pentium75 is right.
upvoted 6 times

MortisG Most Recent 3 weeks, 1 day ago

Selected Answer: A

Correct answer is A. EC2 instance store is not well suited for this scenario since data could be lost if the instance stops or terminates

upvoted 1 times

CloudExpert01 2 months, 1 week ago

Selected Answer: A

Amazon EBS Provisioned IOPS SSD (io2) for the 10 TB of storage needed for high I/O performance video processing.

upvoted 1 times

Mimine87 2 months, 1 week ago

Selected Answer: A

At least 10 TB of storage with max I/O performance (for video processing):
Amazon EBS (Elastic Block Store) provides high-performance block storage for EC2.

Provisioned IOPS SSD (io2) volumes offer the maximum performance and consistency required for intensive workloads like video processing.

□ EC2 instance store is fast but ephemeral — data is lost if the instance stops or fails.

upvoted 1 times

SirDNS 2 months, 3 weeks ago

Selected Answer: D

Can be a bit confusing due to the presence of option A: EBS Provisioned IOPS SSD, Durable = S3, Archive = Glacier, but maximum performance is always equal to instance store. Here they do not need a permanent storage with maximum performance. It will only be used for processing and then sent to a highly durable storage.
D logically cannot be the answer

upvoted 1 times

sk1974 3 months, 3 weeks ago

Selected Answer: A

Instance storage is ephimeral data that gets destroyed when EC2 is stopped/restarted. Unless the question says that data is not important , I will assume that every data that app processed is imporant . So , I will go with A

upvoted 1 times

kyd0nix 4 months, 1 week ago

Selected Answer: D

Ans D.

The requirements for every kind of data is specific. So, you need:
10 TB high I/O / performance = Instance Store (the reqs. doesn't mention anything about durability, just performance)
300 TB very durable = S3
900 TB archival = Glacier

upvoted 1 times

satyaammm 5 months ago

Selected Answer: D

EC2 Instance Store Volumes provide the highest storage performance in AWS and S3 provides durable storage while S3 Glacier is best for archival storage.

upvoted 2 times

DavidPhyo 5 months, 1 week ago

Selected Answer: A

answer is A

upvoted 1 times

Mischi 5 months, 3 weeks ago

Selected Answer: A

The combination of Amazon EBS , Amazon S3 and Amazon S3 Glacier (option A) is the most suitable solution for this media company. It offers:

Maximum I/O performance for video processing with Amazon EBS.
Durability and reliability for media content with Amazon S3.
Cost-effective storage for archived data with Amazon S3 Glacier.
upvoted 2 times

salman7540 5 months, 3 weeks ago

Selected Answer: A

I prefer A over D.

instance storage is not recommended for video processing because it's designed for temporary storage and doesn't provide the durability and persistence needed for video processing. Instead, Amazon EBS is a better choice for video processing because it provides the required performance, durability, and persistence.
Here's some more information about instance storage and Amazon EBS:
Instance storage
A temporary storage volume that acts as a physical hard drive. It's located on disks that are physically attached to the host computer. Instance storage is ideal for temporary storage of information that changes frequently, such as buffers, caches, and scratch data. Data in an instance store persists during the lifetime of its instance, but it's not persistent through instance stops, terminations, or hardware failures.
upvoted 2 times

kimm_10 6 months ago

Selected Answer: B

Why not EC2 instance store?
Instance store is ephemeral (data is lost when the instance stops or terminates).
It lacks durability, making it unsuitable for critical video processing.
upvoted 1 times

0de7d1b 6 months, 3 weeks ago

Selected Answer: A

Explanation:
Amazon EBS (Elastic Block Store) for maximum performance:

Amazon EBS provides high IOPS and low latency storage, which is ideal for video processing workloads that require fast performance.
Amazon S3 for durable data storage:

Amazon S3 is highly durable, scalable, and designed for storing large amounts of data, such as media content (300 TB). It also provides 99.999999999% (11 nines) durability, making it suitable for this requirement.
Amazon S3 Glacier for archival storage:

Amazon S3 Glacier is a cost-effective storage solution for archiving large amounts of data (900 TB) that is infrequently accessed but still needs to be stored securely and durably.
upvoted 3 times

0de7d1b 6 months, 3 weeks ago

C & D. Amazon EC2 instance store for maximum performance: EC2 instance store provides temporary block storage tied to the lifecycle of an instance, so it is not durable or persistent, making it unsuitable for video processing workflows that require consistent data availability
upvoted 1 times

Gizmo2022 7 months, 1 week ago

answer is A
upvoted 1 times

aturret 8 months, 1 week ago

Selected Answer: A

instance store cannot be durable
upvoted 2 times

trainee46 8 months, 3 weeks ago

Selected Answer: D

The requirement ist "maximum performance". Instance storage is more performant than EBS, therefor D must be correct

The requirement ist "maximum performance". Instance storage is more performant than EBS, therefor D must be correct.

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #128 *Topic 1*

A company wants to run applications in containers in the AWS Cloud. These applications are stateless and can tolerate disruptions within the underlying infrastructure. The company needs a solution that minimizes cost and operational overhead.
What should a solutions architect do to meet these requirements?

A. Use Spot Instances in an Amazon EC2 Auto Scaling group to run the application containers.

B. Use Spot Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group. **Most Voted**

C. Use On-Demand Instances in an Amazon EC2 Auto Scaling group to run the application containers.

D. Use On-Demand Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group.

Correct Answer: *B*

Community vote distribution

B (74%) A (25%) D

Comments

bgsanata Highly Voted 2 years, 1 month ago

Selected Answer: A

Requirement is "minimizes cost and operational overhead"
A is better option than B as EKS add additional cost and operational overhead.

upvoted 18 times

Lalo 2 years ago

USING SPOT INSTANCES WITH EKS
https://ec2spotworkshops.com/using_ec2_spot_instances_with_eks.html

upvoted 11 times

Tsige 7 months, 4 weeks ago

Using Amazon EKS with managed node groups simplifies container orchestration by reducing the operational overhead of managing the underlying infrastructure and Kubernetes control plane. EKS automatically handles tasks like patching, scaling, and deploying containers across multiple nodes, further reducing operational effort.
While Spot Instances in an EC2 Auto Scaling group can reduce costs, it requires more manual management of container orchestration (e.g., deploying, scaling, and managing containers across instances), increasing operational overhead compared to EKS.
In the exam its highly recommended to select Managed services Like EKS in this case.
My choice is B.

My choice is B.
upvoted 5 times

19d92c7 7 months, 3 weeks ago

When you use EKS is add operational over head, if ECS was in option then it could have justified here
upvoted 1 times

EMPERBACH 1 year, 1 month ago

Really ???
You want more configuration efforts for container workload on EC2, instead of using EKS ::))
upvoted 7 times

MutiverseAgent 1 year, 11 months ago

In my opinion option A) seems to be a reasonable at first because setting up AWS EKS might be seem as an operation overhead comparing to the option of running the containers inside the EC2 using docker just as you we do on your own machines. However, consider installing docker on multiple EC2 instances and manually manage docker instances and images will end up in chaos, so, as a conclusion, the operational cost of setting up AWS EKS will worth the effort.
upvoted 6 times

GalileoEC2 Highly Voted 2 years, 3 months ago

Answer is A:
Amazon ECS: ECS itself is free, you pay only for Amazon EC2 resources you use.
Amazon EKS: The EKS management layer incurs an additional cost of $144 per month per cluster.
Advantages of Amazon ECS include: Spot instances: Because containers are immutable, you can run many workloads using Amazon EC2 Spot Instances (which can be shut down with no advance notice) and save 90% on on-demand instance costs.
upvoted 10 times

MortisG Most Recent 3 weeks, 1 day ago

Selected Answer: B

Answer is B since it's less operational overhead to run containers on EKS (managed service) than running on EC2 instances which implies installing dockers to orchestrate containers)
upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: A

ai response "Generally, an ECS EC2 Auto Scaling Group is considered cheaper than an EKS Managed Node Group because with ECS, you only pay for the EC2 instances used to run your containers, while EKS adds an additional cost per cluster hour on top of the compute costs for the managed nodes"
upvoted 1 times

Dharmarajan 4 months, 2 weeks ago

Selected Answer: B

Definitely B. running containers on spot EC2 instances means there has to be additional configuration requirements every time an instance comes up. That means making a custom AMI with prepping the spot instance to launch the container = additional operational overhead in maintaining the AMI - with EKS, that overhead is eliminated.
upvoted 2 times

satyaammm 5 months ago

Selected Answer: B

Spot Instances are the cheapest and most suitable here along with EKS for automation and thereby reducing operational overhead.
upvoted 2 times

PaulGa 8 months, 4 weeks ago

Selected Answer: A

Ans A - hint: "stateless" and 'don't care' infrastructure - so Spot and containerisation
upvoted 1 times

huaze_lei 9 months, 1 week ago

Spot instance is definitely the answer for interruptible process. It's between A and B now

Spot instance is definitely the answer for interruptible process. It's between A and B now.
I would reckon Option B requires lesser operational overheads than to maintain own fleet of EC2 servers with containers.
upvoted 1 times

SaurabhTiwari1 9 months, 2 weeks ago

Selected Answer: B

B is correct
upvoted 2 times

ChymKuBoy 11 months, 4 weeks ago

Selected Answer: B

B for sure
upvoted 2 times

Anthony_Rodrigues 1 year ago

IMHO, the option **B** is the right one.
Breaking down the reasons for it:
1 - Spot is much cheaper than on-demand, which already eliminates C&D for cost related.
2 - Even though we can create a bootstrap script to install docker, managing this can be complicated, especially if any of the applications require more than one instance running.
upvoted 4 times

lofzee 1 year ago

Container in ec2 or container on a container platform?
B
upvoted 3 times

EMPERBACH 1 year, 1 month ago

Selected Answer: B

run applications in containers -> Container service not EC2 (no operational overhead to config container workload on EC2)
Spot instance < On-demand cost
upvoted 3 times

vip2 1 year, 3 months ago

Selected Answer: A

Does A implicitly means run spot Instance on ECS?
upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

Spot instances for disruption friendly containers which are also cheaper.
EKS allows using spot instances from a managed node group that takes away the EC2 operational overhead.
Link: https://aws.amazon.com/blogs/containers/amazon-eks-now-supports-provisioning-and-managing-ec2-spot-instances-in-managed-node-groups/
"Previously, customers had to run Spot Instances as self-managed worker nodes in their EKS clusters. This meant doing some heavy lifting such as building and maintaining configuration for Spot Instances in EC2 Auto Scaling groups, deploying a tool for handling Spot interruptions gracefully, deploying AMI updates, and updating the kubelet version running on their worker nodes. Now, all you need to do is supply a single parameter to indicate that a managed node group should launch Spot Instances, and provide multiple instance types that would be used by the underlying EC2 Auto Scaling group."
upvoted 2 times

pipici 1 year, 7 months ago

Selected Answer: A

A has less operational overhead
upvoted 2 times

xdkonorek2 1 year, 7 months ago

Selected Answer: B

running containers without container service like EKS introduce huge operational effort
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #129 *Topic 1*

A company is running a multi-tier web application on premises. The web application is containerized and runs on a number of Linux hosts connected to a PostgreSQL database that contains user records. The operational overhead of maintaining the infrastructure and capacity planning is limiting the company's growth. A solutions architect must improve the application's infrastructure.
Which combination of actions should the solutions architect take to accomplish this? (Choose two.)

A. Migrate the PostgreSQL database to Amazon Aurora. **Most Voted**

B. Migrate the web application to be hosted on Amazon EC2 instances.

C. Set up an Amazon CloudFront distribution for the web application content.

D. Set up Amazon ElastiCache between the web application and the PostgreSQL database.

E. Migrate the web application to be hosted on AWS Fargate with Amazon Elastic Container Service (Amazon ECS). **Most Voted**

Correct Answer: *AE*

Community vote distribution

AE (98%) CE

Comments

ArielSchivo Highly Voted 2 years, 7 months ago

Selected Answer: AE

I would say A and E since Aurora and Fargate are serverless (less operational overhead).
upvoted 14 times

baba365 1 year, 9 months ago

There's a difference between Amazon Aurora and Amazon Aurora Serverless
upvoted 2 times

pentium75 1 year, 5 months ago

"Aurora serverless" is still a flavor of Aurora, it's not a different product.
upvoted 4 times

Manimgh Most Recent 2 months ago

Selected Answer: AE

Aurora and Fargate are the answers

upvoted 1 times

PaulGa 8 months, 4 weeks ago

Selected Answer: AE

Ans A,E -
-PostgreSQL is compatible with Aurora
-Fargate for container service
Both services are serverless

upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: AE

A and E since both Aurora and fargate are serverless.

upvoted 2 times

JTruong 1 year, 5 months ago

Selected Answer: AE

PostgreSQL is compatible w/ Aurora
Fargate & ECS are also paired with containers
A&E

upvoted 2 times

Ruffyit 1 year, 7 months ago

I would say A and E since Aurora and Fargate are serverless (less operational overhead)

upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: AE

Requirement is to reduce operational overhead,
Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication.
AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AE

The reasons are:

Migrating the database to Amazon Aurora provides a high performance, scalable PostgreSQL-compatible database with minimal overhead.
Migrating the containerized web app to Fargate removes the need to provision and manage EC2 instances. Fargate auto-scales.
Together, Aurora and Fargate reduce operational overhead and complexity for the data and application tiers.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: AE

A is the correct answer because migrating the database to Amazon Aurora reduces operational overhead and offers scalability and automated backups.

E is the correct answer because migrating the web application to AWS Fargate with Amazon ECS eliminates the need for infrastructure management, simplifies deployment, and improves resource utilization.

B. Migrating the web application to Amazon EC2 instances would not directly address the operational overhead and capacity planning concerns mentioned in the scenario.

C. Setting up an Amazon CloudFront distribution improves content delivery but does not directly address the operational overhead or capacity planning limitations.